



2020 K-Innovation ODA Program with Vietnam

Policy Consultation on Strengthening the Capacity of Foresight for Science, Technology and Innovation (STI) Strategic Planning in Vietnam 2019-2020

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Policy Consultation on Strengthening the Capacity of Foresight for Science, Technology and Innovation (STI) Strategic Planning in Vietnam 2019-2020

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Executive Summary

The Vietnamese government has emphasized science, technology, and innovation (STI) as the most important keys for implementing the SocioEconomic Development Strategy (SEDS) 2011-2020 plan and realizing sustainable growth. Vietnam has relatively good STI capacity and human resources among ASEAN countries. Unfortunately, however, Vietnam's STI capacity is still lower than that of other developed countries. In this regard, the Vietnamese government needs a strategic approach to STI policies based on scientific methodologies such as technology foresight, given the limited resources and capacity.

Based on the confirmation and acceptance of the project proposal submitted by NISTPASS (currently VISTI), the Korean government affirmed support for Vietnam in making faster progress toward achieving Vietnam's STI development goals through the K-Innovation ODA program "Policy Consultation on Strengthening the Capacity of Foresight for Science, Technology, and Innovation (STI) Strategic Planning in Vietnam 2019-2020."

The Project aims to strengthen the capacity of key agencies and institutions of the Ministry of Science and Technology in the strategic planning for Vietnam's STI. The scope of the Project has been defined based on Vietnam's STI policy priorities and the lessons learned from the Korean experiences in national STI policies and technology foresight. Accordingly, the Project framework includes the following: i) review of Vietnam's national STI strategies and technology foresight experiences; ii) research on the Korean experiences in technology foresight and national STI strategies; iii) local workshops for the diagnosis of Vietnam's technology foresight capacities and for the consensus on the policy consultation's direction and contents; iv) capacity building workshop with technical training programs to enhance the technology foresight capacity of Vietnamese government officials and experts; and v) development of implementation plan for technology foresight by VISTI and review process by STEPI and Korean experts.

At the kick-off meeting for the first year with VISTI, the objectives and scope of the Project were discussed. STEPI and VISTI discussed further specific action plans and agreed on the project programs and activities. During the workshop on national future strategy and technology foresight, the researchers of both counterparts shared insights on Korea's S&T strategic planning and its implications on Vietnam's capacity for national STI policy and technology foresight. Based on the lessons learned from the workshop and the consensus on the Project activities, VISTI and Vietnamese experts are expected to be able to design the framework of the technology foresight exercise.

The research team suggested implications and recommendations on how to develop Vietnam's capacity for national STI strategy and technology foresight. Vietnam is a fast-growing developing country. However, most found that policy considerations are needed to boost productivity and increase competitiveness for sustainable growth. There are also several crucial challenges that should be tackled such as the polarization, transportation problems, and environmental problems that occur during the rapid growth of Vietnam. Among many policy priorities, continuous investment in and political support for STI are important, since it is a major driving force for economic growth. Our findings on Vietnam's policy characteristics indicated that a considerable part of Vietnam's governance adopted the top-down policy design and relatively weak monitoring and evaluation system. Reflecting these characteristics of Vietnam and realizing its future development require establishing national science and technology innovation strategies. Remarkably, technology foresight is an important methodology available in this process. Vietnam should explore its own growth path through the technology foresight. In this regard, it is important to develop Vietnam's own technology foresight methodology by applying the implications from the Korean experience to the Vietnamese environment and characteristics. Limitations as well as policy implications are further discussed.

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CHAPTER 1

Project Overview

Policy Consultation on Strengthening the Capacity of
Foresight for Science, Technology, and Innovation (STI)
Strategic Planning in Vietnam 2019-2020

Chapter 1.

Project Overview

1. Introduction

Following the establishment of diplomatic relations, the range of exchange and cooperation in multiple aspects such as politics, diplomacy, economy, trade, society, and culture increased rapidly.

[Table 1-1] Relationship between Vietnam and South Korea

Major Events and Agreement Status		Key Situations in Each Field After the Establishment of Diplomatic Relations	
1992.12.22.	Started Korea-Vietnam diplomatic relations and set up embassy	Politics	• 2009: Established a strategic partnership • March 2018: Suggested upgrading to “Comprehensive strategic partnership” after summit
1993.05.01	Signed Agreement on Promotion and Protection of Investment (Amended in 2003)		
1994.05.01	Signed Agreement on Double Tax Prevention (Amended in 2014)	Economy	• Three Major Export Markets of Korea: China, US, Vietnam in that order / Four Major Trade Countries of Korea: China, US, Japan, Vietnam in that order) • 2019 Trade Scale: 69.2 billion dollars (Grew by 140 times compared to 1992) • Cumulative investment of 68.3 billion won from 1988 to 2020 to maintain position as the largest investor for Vietnam
2002.12.01	Signed Agreement on a tourism field		
2005.04.01	Signed Agreement on Grant and Technical Cooperation		
2007.06.01	Concluded the KOR-ASEAN FTA		

Major Events and Agreement Status		Key Situations in Each Field After the Establishment of Diplomatic Relations	
2009.10.21	Signed Agreement on Technical Cooperation for Safety and Health		• November 2017: Announced and promoted the “New Southern Policy of South Korea” • June 2019: Korea-Vietnam Meeting with Vuong Dinh Hue Discussed measures for expanding cooperation in 5 fields: trade/investment, finance/development cooperation, infrastructure construction/energy, industry/agriculture/science technology/ICT, and labor/medical health
2015.01.01	Signed Agreement on promoting KOR-Vietnam exchange and cooperation in the agricultural field		
2015.12.20	Concluded the Korea-Vietnam FTA	Culture	• Spread of Korean culture after establishment of diplomatic relations • Spread of Korean culture with dramas from 1990s and 2000s • Spotlight on K-pop • Drama, K-pop, K-beauty, and K-beauty gradually spread on variety show, tourism, books, and education
2017.11.01	Held the 2017 APEC Summit Meeting		
2018.03.23	Held the Korea-Vietnam Summit Meeting		
2019.11.25	ASEAN-Republic of KOREA Commemorative Summit		
2020.04.14	ASEAN+3 Commemorative Summit for COVID-19		

Source: KOTRA & Wikipedia

The Vietnamese government blueprinted becoming one of the industrialized countries by enacting “Socioeconomic Development Strategy (SEDS) 2011-2020” aiming at political and social stability, improvement of international status, and improvement of quality of life.

The Vietnamese government has historically highlighted the priority of science and technology. Notably, science, technology, and innovation are recognized as the most important keys to implementing the SEDS 2011-2020 plan. In this regard, Vietnam has promoted science, technology, and innovation (STI) development in order to achieve sustainable socioeconomic development goals.

Based on the confirmation and acceptance of the project proposal submitted by NISTPASS (currently known as VISTI), the K-Innovation ODA Program has affirmed the commitment to supporting Vietnam in making faster progress toward achieving its STI development goals.

At the same time, the Korean government designated the official development assistance priority areas for Vietnam based on Korea's New Policy for the South and Vietnam's development strategies and development needs as well as Korea's comparative advantage.

Accordingly, the scope of the Project shall be based on the consideration of Vietnam's STI policy priorities and the lessons learned from the previous Korean experiences in national STI policies and technology foresight (Jungwon Lee, et al, 2019).

The aim of the Project is to strengthen the capacity of key agencies and institutions of the Ministry of Science and Technology in the mid- and long-term strategic planning of STI in Vietnam (Jungwon Lee, et al, 2019).

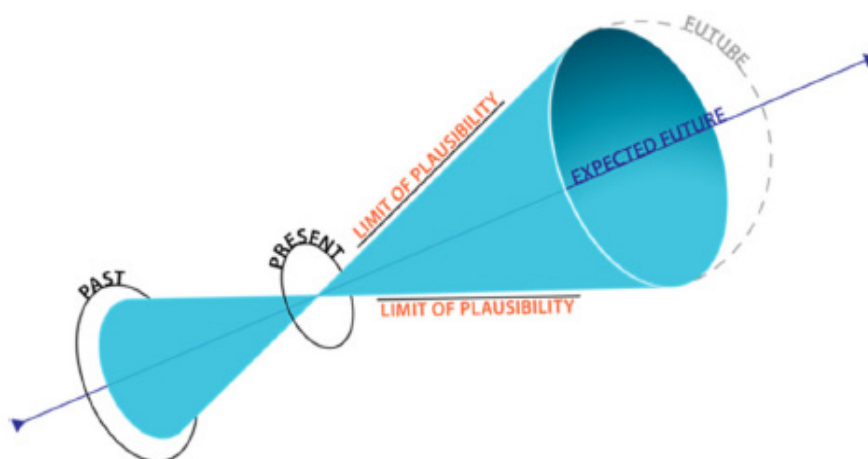
The Science and Technology Policy Institute (STEPI) is a Korean government-sponsored research institute and a leading global think tank that conducts research and analysis on issues pertaining to science, technology, and innovation. STEPI works closely with various ministries, regional governments, and relevant institutes to support policy design. It also has a long experience in providing STI policy consultancy for developing countries, playing the role of conducting STI development cooperation activities with partner countries (Jungwon Lee, et al, 2019).

2. Objectives

The project is titled “Policy Consultation on Strengthening the Capacity for Foresight for Science, Technology, and Innovation (STI) Strategic Planning in Vietnam 2019-2020.” The program duration is from January 1, 2019 to December 31, 2020 (24 months).

The science and technology foresight result shall be utilized in establishing the long-term national economy development strategy of Vietnam to arrange a base for more strategic investment and policies in the science and technology field. Technology foresight refers to the prediction and analysis of possible future technologies in the overall fields of science and technology innovation considering the changes in external and internal environments. As a saying by Lucius A. Seneca goes, “If one does not know to which port one is sailing, no wind is favorable.” Especially, it is necessary to predict the changes in the future strategy, define the future technologies needed, and secure future promising technologies for development in economy, society, and science and technologies (Hyun Im, 2019).

[Figure 1-1] Looking Beyond the Expected Future



Source: Ballinger, M. (2017: p.149)

The purpose of this project is to strengthen the technology foresight competence of related institutes including the Ministry of Science and Technology and Vietnam Institute of Science, Technology, and Innovation for establishing the Vietnam Science and Technology Innovation and Development Strategy (Wangdong Kim, et al, 2019: p.425). Although technology foresight is important, it is hard to solve or predict the future society with simple quantitative index and methods due to the complexity and uncertainty in a cause-and-effect relationship (Jongguk Song, 2009). Thus, in order to enhance the plausibility of future researches and technology foresight, the process of deriving reasonable results through a multi-level utilization of methodologies should be established (Byungwon Park, 2012).

In such context, the policy advisory project seeks to contribute to strengthening the manpower competence of VISTI by transferring various methodologies and promoting workshops.

- 1) The objective of the K-Innovation ODA Project with Vietnam (2019-2020) is to strengthen the capacity of key agencies and institutions of the Ministry of Science and Technology in the mid- and long-term strategic planning of STI in Vietnam.
- 2) The detailed objectives are as follows:
 - Review the current status of STI policy and technology foresight in Vietnam
 - Share Korea's experience in technology foresight
 - Conduct a capacity-building workshop
 - Provide technical advice on the draft report of pilot-scale exercise of a Strategic Plan on Technology Foresight

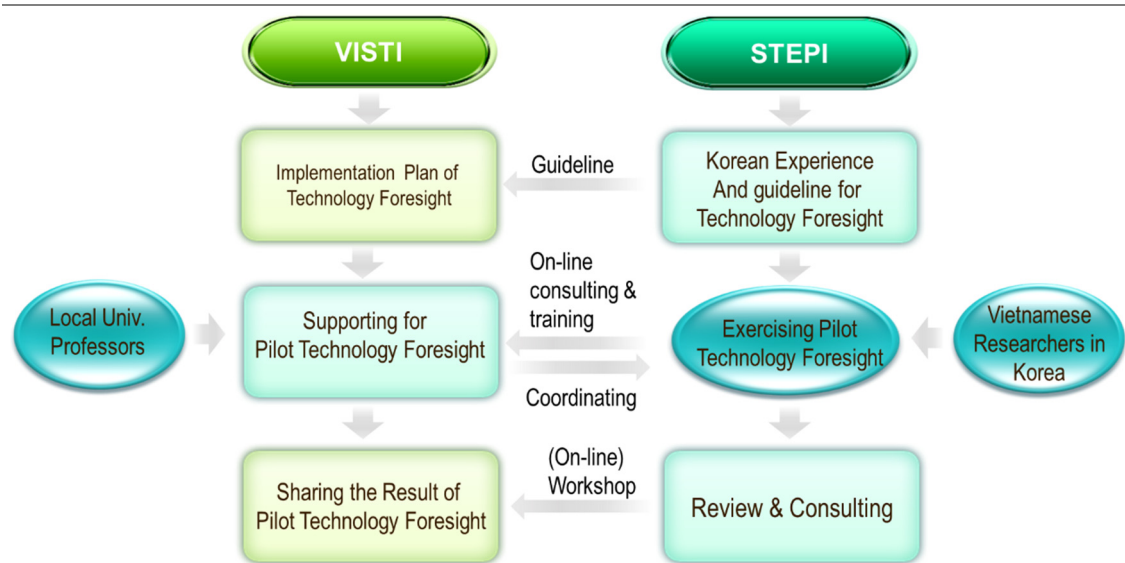
3. Project Framework

3.1 Project Overview

The Project's main activities will be framed as follows:

- (2019) Review on Vietnam's Science and Technology Innovation Development Strategy
- (2020) Support for the Establishment of Technology Foresight in Vietnam
- Review of Vietnam's national STI strategies and technology foresight experiences
- Research on Korean experiences in technology foresight and establishment of national STI strategies or plan based on technology foresight
- Local workshops for the diagnosis of Vietnam's technology foresight capacities and matching work and for the consensus on the direction and contents of policy consultation for major policymakers, government officials, and research institutes of Vietnam
- Capacity-building workshop to enhance the technology foresight capacity of Vietnam's government officials and experts in the STI field through technical training programs
- Development of implementation plan for technology foresight by VISTI and review process by STEPI and Korean experts

[Figure 1-2] Conceptual Framework of the Project



Source: author

3.2 Influence of COVID-19

Since tourism income accounts for 8.4% of the national revenue in Vietnam, COVID-19 has a huge impact on overall society.

According to the Vietnam National Administration of Tourism, 15.5 million foreign tourists visited Vietnam in 2018, with the tourism income (62 trillion dollars) constituting 8.4% of GDP.

Nevertheless, the government of Vietnam enforced a total ban on the entry of foreigners and allowed entry only in exceptional cases starting March 22, 2020 for the safety of its citizens and to prevent the spread of COVID-19.

- ▶ From April 1 to 15, all international flights are prohibited from landing on Vietnamese airports (April 01, 2020) except for special cases.
- ▶ From March 22, entry of all foreigners is prohibited (March 21, 2020)
- ※ Foreigners may enter Vietnam when visiting for △ diplomatic or official affairs △ attending major events (Entry of foreigner is not guaranteed. Entry of foreigner is allowed only after

thorough consultation with Vietnam in advance)

- ※ Suspension of effectiveness of VISA exemption certificate issued to foreigners with Vietnamese citizenship and family members (The effectiveness of the certificate has already been suspended starting March 12, 2020 in the case of Korea)
- ※ Entrants must comply with a medical examination process, and they shall be quarantined. In case of entrants who entered Vietnam for diplomatic purposes or official affairs and special cases (expert, corporate manager, advanced technical manpower), they shall be quarantined in the place where they stay in compliance with the regulation.
- ▶ The certificate for conformity to visa-free entry for 15 days has been temporarily suspended for Koreans (February 29, 2020)
- ▶ VISA issuance has been suspended for all foreigners for 30 days starting March 18, 2020 (March 17, 2020)
- ※ In case of entering Vietnam by △getting a VISA issued for special case (expert, enterpriser, advanced technical manpower) or △ submitting a visa-free certificate or a visa waiver certificate, the entrant shall submit a confirmation of negative result for COVID-19 issued by the authorities and accepted by the government of Vietnam.

Note: The date in parenthesis is the announcement date

Source: Ministry of Foreign Affairs (Overseas Korean Nationals Protection and Crisis Management Division) (2020.4.12.)

All foreigners visiting Vietnam are subject to medical examination and quarantine even for visiting Vietnam for official affairs. In response, the project's on-site modules are converted into contactless ways such as virtual meeting and contact via e-mail.

Due to such actions taken by the government of Vietnam and COVID-19 situations in Korea, the Technology Foresight Capacity Building Workshop scheduled for the first half has been canceled. Likewise, the Competence Strengthening and Training Program with Korea and Local Technology Foresight Performance Spreading Workshop scheduled for the second half are expected to be canceled.

- In response, the original plan for sending a Korean expert to Vietnam to carry out technology foresight is unavailable. As an alternative, there is a discussion on carrying out the pilot-level technology foresight work through the local cooperative institute, VISTI.

4. Project Team

4.1 Research Team

Based on the Project's purposes, a research team was formed consisting of STEPI researchers and three experts. The experts are invited to share their knowledge of Korean development experiences. Vietnam's Technology Foresight Team completes the technology foresight and strategy (plan) establishment tasks based on the technology foresight manual and know-how from Korea. Especially, technology foresight is composed with practical training programs for actual performance (Jungwon Lee, et al, 2019).

The research team consists of three researchers including Dr. Jungwon Lee as Project Manager (PM) and two STI policy experts in STEPI. The research team is responsible for the review on Vietnam's STI strategic plan and technical advice on technology foresight exercise. During the training courses and consultation sessions, the team will provide constructive comments on the pilot exercises of Vietnam and share their evaluation on the Project's progress. External experts have participated in the Project to provide lectures during the workshop and to share technical information or materials for the topics covered. However, the training courses by a Korean expert is converted into the forms of written delivery and virtual meeting due to the influence of COVID-19.

[Table 1-2] Research Team

Name	Organization	Area of expertise	Contact information
Jungwon Lee	STEPI, Senior Research Fellow Project Manager (PM)	STI Policy, Technology Foresight	leejw@stepi.re.kr
Byeongwon Park	STEPI, Research Fellow	Technology Foresight	bwonpark@stepi.re.kr
Minjung Kim	STEPI, Researcher	Project Coordinator	minjungk@stepi.re.kr
Hyun Yim	KISTEP, Senior Research Fellow	Technology Foresight	hyim@kistep.re.kr
Young Il Park	Ewha University, Professor	Strategic Planning	yipark@ewha.ac.kr
Yoo Hyung Won	KIST, Principal Researcher	STI Policy	yhwon@kist.re.kr

Note 1) Last year (2019), Ms. Yong In Choi served as program coordinator.

Note 2) Dr. Byeongwon Park, Dr. Young Il Park, and Dr. Yoo Hyung Won participated in this project in 2019.

4.2 Local Research Team

STEPI collaborated with the Vietnam Institute of Science, Technology, and Innovation (VISTI).

As one of the main institutions of Vietnam in the STI field, VISTI, a policy-oriented research institute, was designated as a counterpart of STEPI for implementing the entire process of the Project. Aside from a researcher from VISTI, Dr. Bach Tan Sinh, two more additional local experts were invited to provide in-depth analysis and related background knowledge of Vietnam's STI policy and Technology Foresight exercises (Jungwon Lee, et al, 2019).

[Table 1-3] Local Research Team

Name	Office	Position	Contact information
Pham Van Hong	VISTI	Deputy Director (Former Director of NISTPASS)	phamvanhong1973@gmail.com
Bach Tan Sinh	-	Head of the Division of Journal and Publication, Department of Science Administration, and Training, VISTI	sinhbt@gmail.com
Ta Doan Trinh	-	Former President of NISTPASS	tadoantrinh1956@gmail.com
Le Dinh Tien	-	Former Deputy Minister of Science and Technology	haletien@yahoo.com.vn

Note: Dr. Bach Tan Sinh was in charge of cooperation with STEPI in 2019 but was replaced by Dr. Pham Van Hong in 2020.

5. Project Activities

5.1 Activity 1: Preparation of the Project

From January to March 2019, the STEPI team conducted a literature review related to the STI strategic planning and technology foresight of Vietnam. Any relevant documents on the issue to enhance understanding should be provided before Activity 2. Detailed agenda for the kick-off meeting and first workshop were developed based on the Project Concept Proposal officially received from the Ministry of Planning and Investment of Vietnam and shared with VISTI for coordination (Jungwon Lee, et al, 2019).

5.2 Activity 2: Kick-off Meeting and First Workshop

The STEPI team visited Hanoi, Vietnam on April 15 to 19, 2019 for the kick-off meeting and the first workshop. Specifically, the workshop aimed to share Korea's experiences in the identified topics, including national strategic planning for STI, introduction of technology foresight, and strategic planning of Government Research Institutes. The Vietnamese counterpart also gave presentations on the STI Policy of Vietnam and technology foresight in Vietnam in order to promote in-depth understanding of Vietnam's current status and experiences in foresight. Afterward, STEPI and VISTI could build a consensus on the expected results and outcome (Jungwon Lee, et al, 2019).

In 2020, the STEPI team was also going to visit Hanoi, Vietnam in April for the kick-off meeting and the first workshop, but it was impossible due to COVID-19. The workshop would be held to share the experiences of Korea on the technology foresight. STEPI and VISTI would also discuss the implementation of the Pilot Technology Foresight in Vietnam, for which the contents and process including scope, time span, and implementation process would be discussed.

5.3 Activity 3: Capacity-building Workshop in Korea

The capacity-building workshop was going to be held in 2020 to share the best practices of Korea and to practice technology foresight as an implementation strategy of the STI Policy, but it was impossible due to COVID-19. In the original plan, a delegation of 10, more or less, from VISTI and relevant institutions would be invited to Korea. The workshop would be organized to share the best practices of Korea and to practice technology foresight. If necessary, an interactive workshop on the development of pilot technology foresight would be organized. They might also visit the relevant Korean organizations and sites.

5.4 Activity 4: Policy Recommendation on the Draft of Pilot Technology Foresight of Vietnam

The draft of the pilot technology foresight in Vietnam will be reviewed by the research team for progress assessment.

5.5 Activity 5: Dissemination Workshop

A final workshop is expected to be held in December 2020. During the workshop, the STEPI team explains the results of the pilot technology foresight and discusses with Vietnamese stakeholders. Stakeholders from the public and private sector, academe, and research institutes involved in STI policies will be invited to the event, which will disseminate the details of the program. The STEPI research team will deliver comments on the draft policies, and the final report of the consultation will be distributed. Additionally, evaluation interview is to be conducted to assess the relevance and sustainability of recommendations and to build upon possible follow-up programs (need to discuss the COVID-19 situation).

6. Project Schedule

The following is the overall schedule of the project:

[Table 1-4] Project Schedule

Activities	Jan.	Feb.	Mar.	Apr.	May	Jun.	Jul.	Aug.	Sept.	Oct.	Nov.	Dec.
Administrative Procedure in the Government of Vietnam ^{Note 1)}												
Consultation on 2 nd -Year Plan and Schedule ^{Note 2)}												
Preparation of Technology Foresight Manual												
Interim Report												
Competence Strengthening Training												
Dissemination Workshop												
Final Report												

Note 1) Administrative procedures necessary for the ODA project of the government of Vietnam

Note 2) Additional consultation depending on the COVID-19 situation

※ The plan has been modified due to COVID-19 but may change depending on the development of the situation.

7. Project Outputs

STEPI and VISTI will draw up the following deliverables:

- **Deliverable 1:** Project Proposal and Action Plan (STEPI)
- **Deliverable 2:** Program Agenda for Kick-off Meeting and Workshop (STEPI)
- **Deliverable 3:** Guidelines for the Pilot Technology Foresight in Vietnam (STEPI)
- **Deliverable 4:** Report on Pilot-scale Technology Foresight (STEPI)
- **Deliverable 5:** Final Report of a Strategic Plan on Technology Foresight (STEPI)



CHAPTER 2

STI Policy and Technology Foresight in Vietnam

Policy Consultation on Strengthening the Capacity of
Foresight for Science, Technology, and Innovation (STI)
Strategic Planning in Vietnam 2019-2020



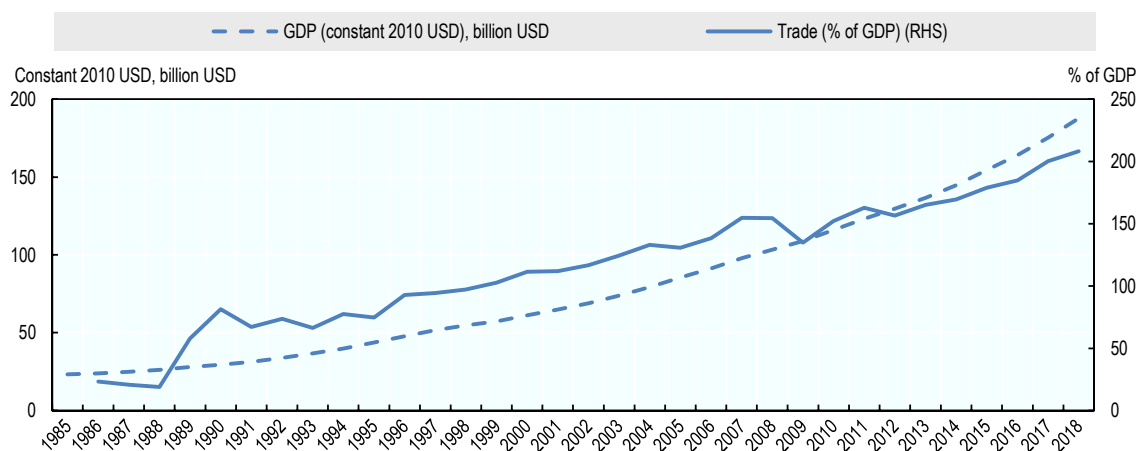
Chapter 2. STI Policy and Technology Foresight in Vietnam

1. Vietnam's STI Status

Since the 1990s, the most striking feature of the economy of Vietnam has been its remarkable economic growth. Vietnam has been regarded as an emerging East Asian tiger. The country's average GDP growth was percent per year from 2000 to 2015. Vietnam recorded GDP growth of 7.08 % in 2018, the peak since 2011 and well above the projected growth rate in 2018 (6.8 percent) (Cameron, A., et al, 2019). In the context of development circles since the early 2000s, the country's average GDP growth was percent per year from 2000 to 2015.

However, several leadership scholars have argued that this momentum could not be sustained without a sizable increase in the role of government pushing for science and technology, including sufficient investment for science and engineering education and support for public and private laboratories (Klingler-Vidra, R. & Wade, R., 2020; World Bank, 2014).

[Figure 2-1] Vietnam's GDP Trends



Source: OECD, 2020b.

Vietnam's GDP growth has been slowing in a less buoyant international environment. Previous sources of growth are diminishing in power, raising the threat of a "middle-income trap." Vietnam will have to rely more on productivity gains driven by innovation. This will require considerable improvements in domestic innovation capabilities (OECD, 2020a).

Taking a look at Vietnam's competence in science and technology innovation as stated in the Global Innovation Index (GII), the enterprise accounts for a high proportion (64.1%) of the total R&D expense. Likewise, the high proportion of hi-tech imports and high proportion of FDI maintenance show active efforts to absorb advanced knowledge from outside.

Due to the low proportion of knowledge-intensive workers and low corporate investment in education and training, however, it is important for Vietnam to cultivate excellent science and technology manpower.

Among developing countries, Vietnam is highly interested in science and technology innovation, and it is currently expanding investment in science and technology following the recent economic development. However, Vietnam still lacks competence in science and technology, and its investment scale is limited. Therefore, it is essential to promote strategic science and technology policies and predict technologies as a strategic tool.

Against such background, the Vietnam Institute of Science, Technology, and Innovation

(VISTI) requested the Science and Technology Policy Institute (STEPI) for a policy advisory project to strengthen the competence of manpower who will be in charge of “Vietnam’s Technology Foresight and Strategy Plan Establishment.”

[Table 2-1] Vietnam’s Global Innovation Index (GII 2019)

Global Innovation Index	Score	Rank
• R&D	7.4	67
- Researchers, FTE/mil. pop	700.8	58
- Gross expenditure on R&D, % GDP	0.5	61
- Global R&D companies, avg. exp. top 3, mil. US\$	0	43
• ICTs	57.5	82
- ICT access	48.8	90
- ICT use	38.7	92
- Government’s online service	73.6	57
• Knowledge workers	22.8	102
- Knowledge-intensive employment, %	1.1	117
- GERD performed by business, % GDP	0.4	42
- GERD financed by business, %	64.1	8
• Innovation linkages	20.0	86
- University/Industry research collaboration	38.6	75
- State of cluster development	45.2	74
- GERD financed by foreign source, %	4.5	64
- JV strategic alliance deals/bil. PPP\$ GDP	0.0	49
• Knowledge absorption	47.12	23
- Hi-tech imports, % total trade	26.5	1
- ICT services imports, % total trade	0.0	126
- FDI net inflows, % GDP	6.2	23
• Knowledge creation	8.1	80
- Patents by origin/bil. PPP\$ GDP	0.9	65
- Scientific & technical articles/bil. PPP\$ GDP	5.6	74
• Knowledge diffusion	42.1	18
- Hi-tech net exports, % total trade	32.9	1
- ICT services exports, % total trade	0.1	125
- FDI net outflows, % GDP	0.4	71

Source: Global Innovation Index 2020, <https://bit.ly/3jQzea6> (accessed on November 2, 2020)

2. Vietnam's National STI System

2.1 Science and Technology Policy in Vietnam

Governments play a key role in providing long-term orientation on social and economic priorities, ensuring that resources for innovation are adequate, public actors perform well, and various components of the innovation system link up and form a coherent whole. Innovation system governance in Vietnam has been plagued by a number of shortcomings that can be attributed to lack of effective commitment, strategic coordination, and implementation of government policies (OECD, 2020a).

[Table 2-2] Vietnam's Key STI Policy (1981–2018)

-
- 1981: Decision 175 allows the (private) signing of R&D contracts.
 - 1987: Decision 134 encourages science and technology.
 - 1988: Ordinance on Foreign Technology Transfer provides for the implementation of the 1987 Law on Foreign Investment.
 - 1992: Decree 35 allows the establishment of non-state R&D organizations.
 - 1993: National Centre for Scientific Research renamed the National Centre for Natural Science and Technology, with the aim of conducting fundamental and applied research.
 - 1996: Resolution 2 names S&T development strategy as part of the industrialization and modernization mission.
 - 1999: Decree 119 sets the legal basis for public support for S&T firms.
 - 2000: First Law on S&T; Decision 850 creates national laboratories.
 - 2003: Announcement that the first S&T Development Strategy will be set by 2010, and the National Foundation for Science and Technology Development (NAFOSTED) is founded.
 - 2005: Decision 214 on the “development of technology market.”
 - 2006: Law on technology transfer and Prime Minister Decree 80 on commercialization.
 - 2007: Creation of the State Agency for Technology and Innovation (SATI) and the National Council for Science and Technology Policy to advise the Prime Minister.
 - 2008: High-technology Law passed; NAFOSTED and Science and Technology Fund launch operations.
 - 2011: Plan to foster international integration in S&T and creation of the National Agency for Technology Entrepreneurship and Commercialisation Development (NATEC).
-

-
- 2012: The Prime Minister approves the 2011–2020 Science and Technology Development Strategy and Decision.
 - No. 592 sets a goal of 3,000 S&T SMEs by 2015 and 5,000 by 2020.
 - 2013: The S&T Law recognizes S&T enterprises for incentives and R&D access; Central Resolution 6 advocates
 - Development of S&T in the context of Socialist-oriented market economy and international integration; MoST initiates the Vietnam Silicon Valley Project.
 - 2015: TechFest Vietnam, an annual event for the startup ecosystem, and National Technology Innovation Fund, both launched by MoST.
 - 2016: Prime Minister Decision 844 to support the national startup ecosystem until 2025.
 - 2017: The SME Support Law offers tax incentives, credit access, and accounting support.
 - 2018: The Law on Supporting Small and Medium-Sized Enterprises and Decree 38 on innovative startup investments came into effect.
-

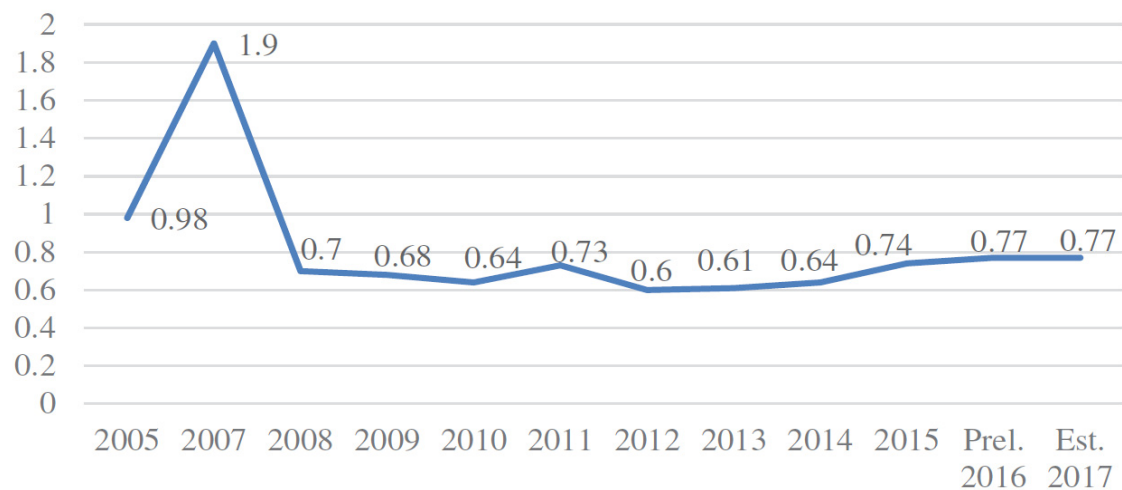
Source: Klingler-Vidra, R., & Wade, R. (2020: p.5)

The table above illustrates how policy and institutional attempts to stimulate technology upgrade were made over time. Notice that the emphasis throughout has been on policies to promote strong private cooperation, which simply aim to augment the number of firms, as distinguished from initiating advancements in strategic orientation (Klingler-Vidra, R., & Wade, R., 2020).

2.2 STI System of Vietnam

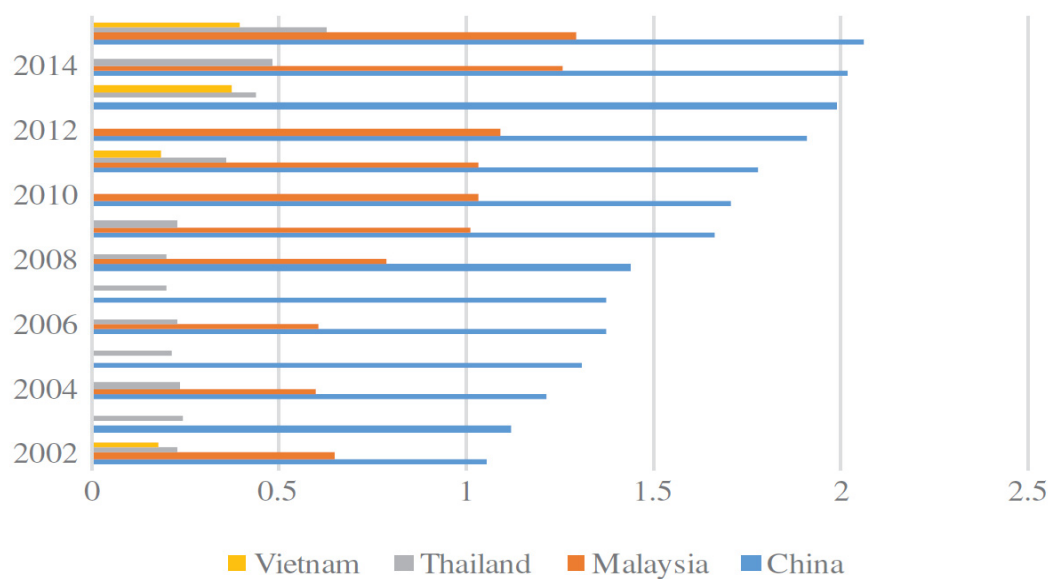
Notably, the low growth of total R&D expenditure has long been attributed to the significantly low investment based on public spending on R&D. Vietnam has advanced the legal basis for STI and established several new institutions engaged in steering and funding R&D. Still, progress in building a modern institutional framework has to continue in a timely fashion. Professionalized agencies with sufficient degree of operational autonomy and larger portfolios can improve the process of policy implementation. Most found that implementation capacity is a key success factor for East Asian countries. This implies that there is an urgent need to strengthen the information base for STI policy, indicators, and evaluation practices (OECD, 2020a).

[Figure 2-2] Government S&T Expenditure as a Percentage of State Budget



Source: Klingler-Vidra, R., & Wade, R. (2020: p.6)

[Figure 2-3] R&D Intensity (share of GDP)



Source: Word Bank; Klingler-Vidra, R., & Wade, R. (2020: p.7)

3.

Overview of Vietnam's Overall Transportation Status

a) Overview of Vietnam's overall transportation status

Due to poor public transportation and road infrastructure, there is high preference for motorcycle as a means of transportation.

As of 2019, the number of automobiles was limited to 3 million. On the other hand, motorcycles numbered 58.17 million, recording a high ratio of 604 motorcycles per 1,000 population.

[Table 2-3] Current Status of Transportation Usage in Vietnam

Classification	2008	2009	2010	2011	2012	2013	2014	2015	2016	2017
Bus	97	104	98	102.5	106.9	111	112.5	118.4	136.7	154
Truck	398	476	552	607	652	696	752	858	1,109	1,138
Car	947	1,138	1,274	1,463	1,590	1,669	1,837	2,107	2,516	2,902
Carriage	384	484	557	672	727	799	900	1,033	1,270	1,495
Motorcycle	25,273	28,195	31,155	33,774	36,894	38,643	41,197	44,128	47,131	54,063

Unit: 1,000

Source: ASEAN Stats 2018

Note: Total number of registered buses (thousand), Number of registered trucks (thousand), Total number of registered road motor vehicles (thousand), Number of registered passenger cars (thousand), Total number of registered motorcycles (thousand)

As for the status of road infrastructure in Vietnam, given its geological conditions, approximately 40% of national roads are located on mountainous areas. On top of that, proper transportation infrastructure has not been built due to limited budget and geographical problems. Roads are classified into 7 types including highway and national road. Total road length is 346,498Km.

[Table 2-4] Status of Road Infrastructure in Vietnam

Status of Highway Routes of Vietnam	Number of Routes and Route Length of Vietnam			
	Type of Road	No. of Route	Road Length (Km)	Percentage (%)
	Highway	16	906	8.1
	National Road	146	23,818	6.9
	Local Road	998	28,167	8.1
	Zone Road	868	60,972	17.4
	Agricultural and Fishing Village Road	61,402	177,024	51.1
	Town Road	25,495	45,377	13.1
	Special Road	-	10,838	3.1
	Total		346,498	100

Source: Youngin Kwon et al (2019)

Demand on Road Traffic	Development Goal
- No. of Passengers: 5.5 billion - Volume of Goods Transported: 760 million tons - No. of Vehicles: 2.8 ~3 million	- Construction of 24 highways with total length of 2,381km (partially including the highway section) - Improvement of all national roads to satisfy the national standard - Securing durability for all bridges - Asphalt pavement for all roads managed by the local governments - Utilization of road infrastructure facility area on 16~26% of construction area in the city

Source: Youngin Kwon, et al (2019)

b) Current status of transportation in Hanoi

Hanoi is the capital of Vietnam and is located at the center of Red River Delta, a granary near the Chinese border and Laos border in the northern part of Vietnam. The city holds a long history, and it has been serving as the central city of North Vietnam and Red River Delta.

The city has an annual population growth rate of 3.5 %, and the primacy of Ho Chi Minh City is gradually getting better.

[Table 2-5] Status of Transportation Sharing in Hanoi

Mode	Trips per Year (Million)
Public Transport	562.80
Scheduled bus and minibus	432.80
Bus	130.0
Train, metro, train	0
Ferry	0
Active Transport	241.77
Walking	113.33
Bicycle	128.44
Individual Motorized Vehicles (taxi, passenger car, motorcycle and others, i.e., trucks, etc.)	6,750.92
Total	7,555.49
Public and Active	804.57
Mode Share of Active and Public Transport (%)	10.65



Source: Chung (2019)

Private vehicles and motorcycles make up the biggest chunk of Hanoi's transportation system. According to the research conducted by the General Statistics Office of Vietnam in 2017, there are 485,955 cars and 5,255,245 motorcycles. Moreover, there are 160 cars and 2,561 motorcycles, both per 10kg. These statistics suggest that motorcycles in Vietnam have a very high utilization rate. According to Hanoi's Department of Transportation, taxis, passenger cars, and motorcycles account for 90% of total traffic, with non-motorized transportation means such as bicycle taking up 10%.

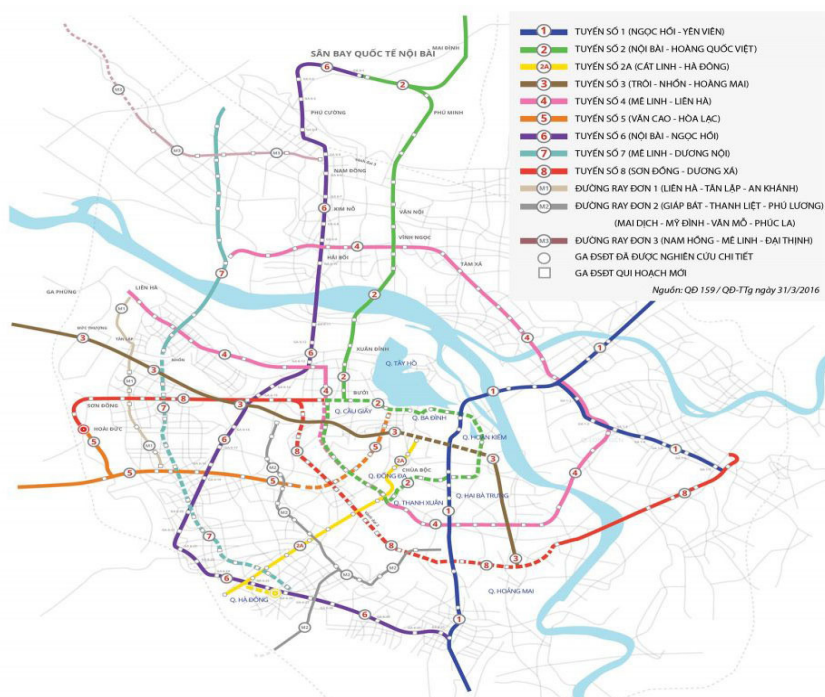
A public transportation-centered traffic policy is promoted to solve the problem of congested areas caused by unclear lanes and lack of separation between transportation means. From 2011 to 2016, road length increased by 3.9% per year, and road area increased by 0.25% annually. As of 2017, there are 1,147 roads in Hanoi with total road length of 2,052km and road area of 16.1 km². Since the early 2000s, Hanoi has been focusing on securing bus routes, and it runs 1,599 buses on 123 routes as of 2018. However, the numbers of buses and routes are not enough to satisfy the demand. Although Hanoi is extending subways, there are a few issues such as changes in the surrounding environment and frequent delay in opening the subway.

Hanoi is the starting point/origin of Vietnam railway lines. The North-South Railway runs from Hanoi to Ho Chi Minh City and connects to other northern cities such as Hai Phong. Hanoi runs international trains that head toward Kunming and Nanning in China. Hanoi is actively promoting the subway plan for solving urban traffic issues by designing the "2020 Hanoi Traffic Planning" in 2003 and approving the "2030 Hanoi Traffic Transportation Planning" in 2018.

[Table 2-6] Status of Hanoi's Metro

2A Line Bridge	Map of Routes Currently Under Construction
	

Hanoi Metro Plan from the 2030 Hanoi Traffic Transportation Planning



Source: KOTRA (2016.07)



CHAPTER 3

Sharing Korea's Experiences in Science and Technology Planning and Technology Foresight

Policy Consultation on Strengthening the Capacity of
Foresight for Science, Technology, and Innovation (STI)
Strategic Planning in Vietnam 2019-2020

Chapter 3. Sharing Korea's Experiences in Science and Technology Planning and Technology Foresight

1. Technology Foresight in Korea

1.1 History of Korea's Technology Foresight¹

According to Korea's Science and Technology Basic Law, Korea conducts technology foresight every five years. Implemented by the Korea Institute of S&T Evaluation and Planning (KISTEP), the project runs for one and a half years. The Korean technology foresight aims to provide the vision and direction of emerging S&T areas by identifying new technologies that may have high potential in terms of growth of national wealth and betterment of quality of human life (Jungwon Lee, et al, 2019).

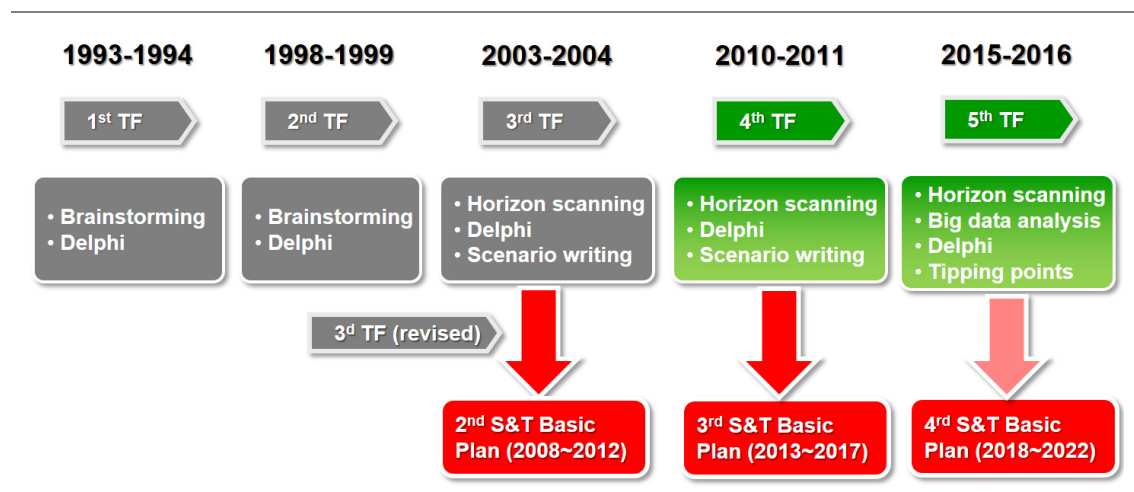
Based on the technology foresight, Korea develops a new S&T Basic Plan every five years alongside the launch of a new administration. All Science and Technology planning activities at the national level are linked to the National Science and Technology Basic Plan. The National Mid- and Long-term Science and Technology Development Strategy selects the national core strategic technologies based on the TF results on future technologies. Based on this, the government establishes a strategic technology roadmap. The technological development of the initiatives being focused on in the S&T Basic Plan reflects it. The 652 future technologies identified by the fourth TF went through a review process by committees responsible for the national R&D budget as well as by R&D-related ministries (Gokhberg, L., et al, 2016). As a result, a list of 120 national strategic technologies was compiled; these technologies were also identified by the third S&T Basic Plan (Choi M., Choi H., 2015). The importance attributed to Korea's technology foresight (TFs) has grown with

¹ Choi M., Choi H. (2015)

every round of TF (Choi M., Choi H., 2015; Jungwon Lee, et al, 2019).

Since the first TF for the period 1993–1994, Korean TF has been further developed and advanced in response to the socioeconomic changes and increasing demand for TF exercises. Since the Framework Act on Science and Technology in 2001, which stipulated carrying out TFs regularly, national TFs have been conducted every five years. Thus, from the third TF onward, Korea has been carrying out TFs every five years. In 2007, the third TF was revised to increase complementarities with the S&T Basic Plan, the nation’s top-level plan in the field of S&T. The results of the revised TF were directly reflected to the second S&T Basic Plan. Furthermore, the results of the fourth TF (conducted during the period 2010–2011) found expression in the third S&T Basic Plan. All four TFs performed to date have primarily used the Delphi method (Gokhberg, L., et al, 2016). Beginning with the third TF, future social trends were first identified, and then future technologies were predicted based on these trends (Gokhberg, L., et al, 2016); moreover, scenarios were developed based on the results of the TF. Currently, the Ministry of Science, ICT, and Future Planning (MSIP) is responsible for TFs, with KISTEP conducting the TFs (Choi M., Choi H., 2015; Jungwon Lee, et al, 2019).

[Figure 3-1] History of Korea's Technology Foresight



Source: Hyun Yim (2019)

2. Pilot Project for Vietnam's Technology Foresight

2.1 Overview

a) Background

STEPI has prepared to collaborate with VISTI for the pilot project for Technology Foresight in Vietnam. We conducted a pilot project for technology foresight with experts staying in Korea and Vietnam.

- Period: July ~ November 2020
- Purpose: To deliver technology foresight methodologies by promoting Vietnam's technology foresight pilot project

The main items are as follows:

- To set areas of the technology foresight pilot project
- To select subjects of the technology foresight survey
- To analyze the result of the technology foresight survey

b) Target areas: future transportation

The analysis of vehicle and transportation status showed the need for technology foresight to solve the social issues caused by lack of public transportation and road infrastructures in Vietnam.

As the project is a demonstrative project, the project scope has been limited to technology foresight on the future transportation of Hanoi instead of all Vietnamese regions to complete the project within the period.

2.2 Survey of Technical Prediction of Hanoi

a) Survey overview

The purpose of this survey is to provide fundamental data and implications to improve traffic congestion and issues related to traffic, establish government policies to improve mobility in the future, and reinforce technical prediction and capability in Vietnam.

- Survey period: Web survey is conducted from October 15 to 18, 2020 (about 1 week).
- Survey subjects: A survey is conducted among Vietnamese international students or experts residing in Korea such as those belonging to the Science & Technology Policy Institute or local experts in Vietnam.
- Collect emails from local experts in Vietnam through CV in the teaching staff homepage in the field of traffic in universities in Vietnam or corresponding authors of papers related to “Vietnam Transportation.”
- Survey method: Prepare web survey questions and send survey notice and survey link via email.

[Table 3-1] Survey Method

Classification	Contents
Web survey	Send notice and survey contents via email to the survey population

b) Survey contents

Configure the survey questions related to traffic in large cities in Vietnam based on 13 promising technologies related to transportation in Vietnam in the future.

- Choose the final 13 promising technologies among 18 related to traffic in Vietnam in the future based on a review by experts.

- Prepare questions including the severity of each issue of traffic, extent of effect of each policy, importance of each promising technology, and period of realization.

[Table 3-2] Survey Contents

Classification	Contents
Basic information	• Basic information about the respondents (email, gender, age, occupation, highest educational attainment, etc.)
Severity of traffic issues and policy effect	• Evaluate the severity of each traffic issue in large cities in Vietnam • Evaluate the effect of each policy for solving traffic issues
Promising technology in the future in each area	• Importance of promising technology in 3 areas and common areas • Technical competitiveness in Vietnam • Period of realizing the technology in Vietnam and the world • Core research topics • Common policies needed to realize technology

c) Survey results

1. Recognition of Traffic in Large Cities in Vietnam and Evaluation of Policy Effect

■ Severity of each traffic issue

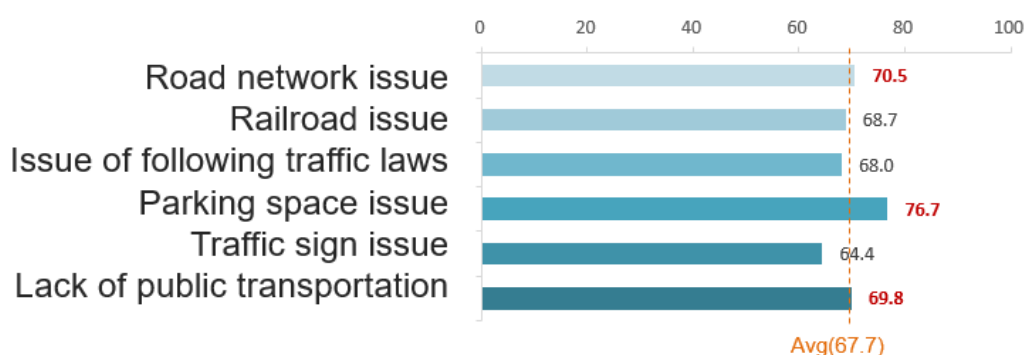
According to the results of investigation on the severity of each traffic issue in large cities in Vietnam, “parking space issue” got the highest score with 76.7, followed by “road network issue” with 70.6 and “lack of public transportation” with 69.8.

[Table 3-3] Severity Level of Traffic Issue in Large Cities in Vietnam

Classification	Road network issue		Railroad issue		Issue of following traffic laws		Parking space issue		Traffic sign issue		Lack of public transportation	
	Case	Proportion	Case	Proportion	Case	Proportion	Case	Proportion	Case	Proportion	Case	Proportion
Very low	3	5%	5	9%	2	4%	5	9%	2	4%	1	2%
Low	8	15%	9	16%	10	18%	6	11%	12	22%	10	18%
Neutral	16	29%	13	24%	16	29%	4	7%	21	38%	16	29%
High	13	24%	13	24%	18	33%	18	33%	12	22%	17	31%
Very high	15	27%	15	27%	9	16%	22	40%	8	15%	11	20%
Total	55	100%	55	100%	55	100%	55	100%	55	100%	55	100%
Score	70.5		68.7		68.0		76.7		64.4		69.8	

* Score was calculated as very high =100, high=80, neutral=60, low=40, and very low=20.

[Figure 3-2] Severity Level of Traffic Issue in Large Cities in Vietnam



■ Effect of policies related to traffic issues

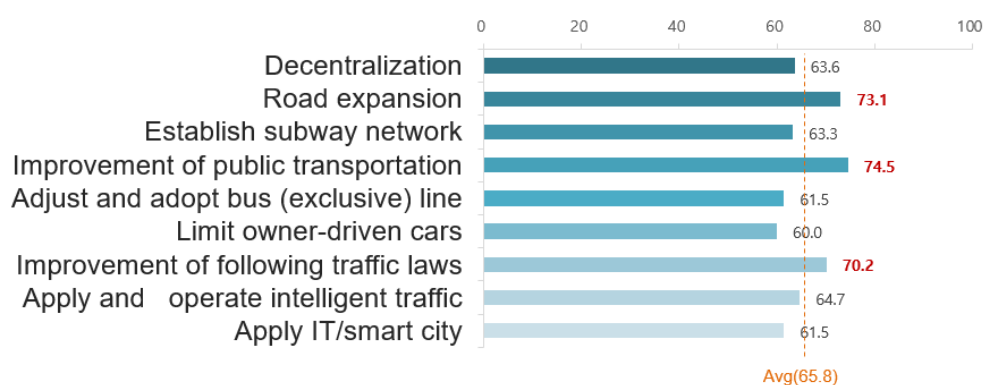
According to the results of investigation on the effect of each policy to resolve the traffic issue, “improvement of public transportation” got the highest score with 74.5, followed by “expansion of road” with 73.1 and “improvement of following traffic laws” with 70.2.

[Table 3-4] Effect of Policies for Traffic Issues

Classification	Decentralization		Road expansion		Establish subway network		Improvement of public transportation		Adjust and adopt bus (exclusive) line		Limit owner-driven cars		Improvement of following traffic laws		Apply and operate intelligent traffic		Apply IT/smart city	
	Case	Proportion	Case	Proportion	Case	Proportion	Case	Proportion	Case	Proportion		Proportion	Case	Proportion	Case	Proportion	Case	Proportion
Very low	3	5%	1	2%	4	7%	1	2%	3	5%	1	2%	0	0%	5	9%	8	15%
Low	11	20%	4	7%	9	16%	6	11%	18	33%	20	36%	8	15%	11	20%	11	20%
Neutral	22	40%	17	31%	21	38%	16	29%	13	24%	19	35%	21	38%	15	27%	15	27%
High	11	20%	24	44%	16	29%	16	29%	14	25%	8	15%	16	29%	14	25%	11	20%
Very high	8	15%	9	16%	5	9%	16	29%	7	13%	7	13%	10	18%	10	18%	10	18%
Total	55	100%	55	100%	55	100%	55	100%	55	100%	55	100%	55	100%	55	100%	55	100%
Score	63.6		73.1		63.3		74.5		61.5		60.0		70.2		64.7		61.5	

* Score was calculated as very high =100, high=80, neutral=60, low=40, and very low=20.

[Figure 3-3] Effect of Policies for Traffic Issues



2.3 Evaluation of Promising Technologies Related to Traffic in Large Cities in Vietnam in the Future

a) Comparison and analysis of 13 promising technologies

■ Importance of promising technology

According to the results of the investigation on the importance in each field in Vietnamese society in the future, the railroad traffic field got the highest score (and regarded as the most important) with 83.3 on average, followed by road traffic field with 81.3 and aviation traffic field with 73.3.

According to promising technologies evaluated to be highly important in each area, “AI traffic operation system” (82.7) was perceived to be highly important in the field of road traffic, followed by “highly efficient hybrid railroad vehicle” (86.1) in the field of railroad traffic, “autonomous flying support infrastructure” and “manned or unmanned autonomous aircraft designing, manufacturing, and certifying system” (73.3 each) in the aviation traffic field, and “traffic infrastructure rapidly improving or replacing devices” (81.7) in the common area.

[Table 3-5] Importance of Each Promising Technology in the Future

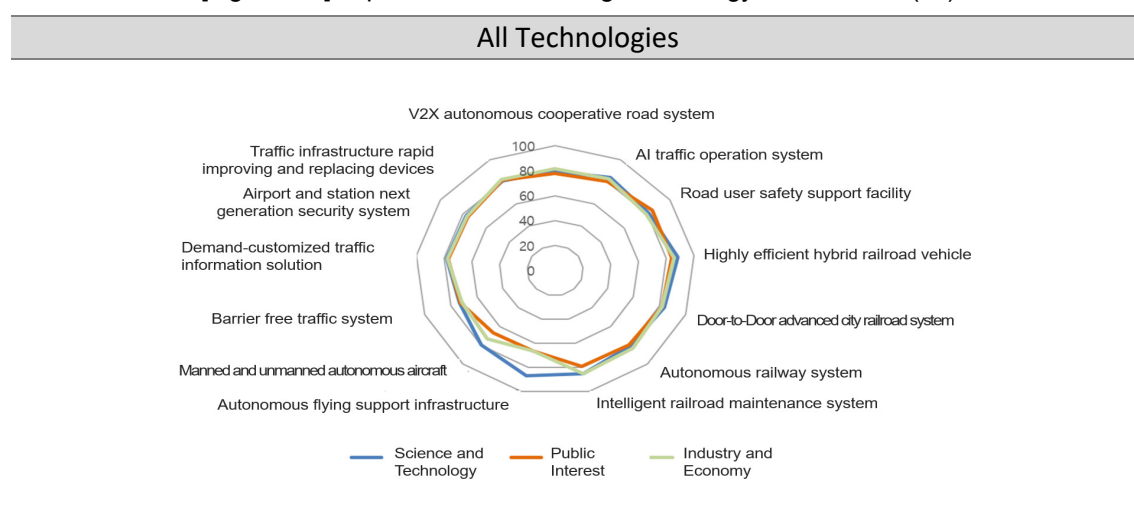
Classification		Scientific technology part	Public benefit	Industrial and economic aspects	Average
Road traffic	V2X autonomous cooperative road system	79.3	77.9	81.4	79.5
	AI traffic operation system	84.4	80.7	83.0	82.7
	Road user safety support facility	81.5	84.6	79.2	81.8
	Subtotal (average)	81.8	81.1	81.2	81.3
Railroad traffic	Highly efficient hybrid railroad vehicle	88.3	84.2	85.8	86.1
	Door-to-door advanced city railroad system	83.5	81.7	81.7	82.3
	Autonomous railway system	80.9	80.0	83.5	81.4
	Intelligent railroad maintenance system	85.2	79.1	85.2	83.2

Chapter 3. Sharing Korea's Experiences in Science and Technology Planning and the Technology Foresight

Classification		Scientific technology part	Public benefit	Industrial and economic aspects	Average
	Subtotal (average)	84.5	81.3	84.1	83.3
Aviation traffic	Autonomous flying support infrastructure	86.7	66.7	66.7	73.3
	Manned and unmanned autonomous aircraft designing, manufacturing, and certifying system	80.0	70.0	70.0	73.3
	Subtotal (average)	83.3	66.7	70.0	73.3
Common	Barrier-free traffic system	73.0	72.4	71.4	72.3
	Demand-customized traffic information solution	77.9	76.9	77.4	77.4
	Airport and station next-generation security system	77.4	75.9	76.4	76.6
	Traffic infrastructure rapidly improving and replacing devices	81.0	81.5	82.6	81.7
	Subtotal (average)	77.3	76.7	76.9	77.0
Total (average)		81.5	77.6	79.1	79.4

※ Score was calculated as very high =100, high=80, neutral=60, low=40, and very low=20.

[Figure 3-4] Importance of Promising Technology in the Future (All)



■ Technical competitiveness

According to the results of investigation on the technical competitiveness in Vietnam with regard to promising technologies in the future, the road traffic area got the highest score (and regarded as the most competitive) with 65.4, followed by the railroad traffic area with 64.7 and the aviation traffic area with 60.0.

According to the technical competitiveness in each of the 13 promising technologies, “demand-customized traffic information solution” got the highest score with 68.2 followed by “airport and station next-generation security system” and “traffic infrastructure rapidly improving and replacing devices” with 66.7 each.

[Table 3-6] Technical Competitiveness in Each Promising Technology in the Future

Areas	Classification	Average Score	
		Each Technology	Each Field
Road Traffic	V2X autonomous cooperative road system	66.4	65.4
	AI traffic operation system	63.7	
	Road user safety support facility	66.2	
Railroad Traffic	Highly efficient hybrid railroad vehicle	65.8	64.7
	Door-to-door advanced city railroad system	66.1	
	Autonomous railway system	60.0	
	Intelligent railroad maintenance system	67.0	
Aviation Traffic	Autonomous flying support infrastructure	60.0	63.3
	Manned and unmanned autonomous aircraft designing, manufacturing, and certifying system	66.7	
Common	Barrier-free traffic system	64.3	66.7
	Demand-customized traffic information solution	68.2	
	Airport and station next-generation security system	67.7	
	Traffic infrastructure rapidly improving and replacing devices	66.7	
Average		65.3	

※ Score was calculated as very high =100, high=80, neutral=60, low=40, and very low=20.

■ Period of realizing technology in the future

For the questions on the period of realizing technology in the future in Vietnam, there was more than 1 response as to how it was impossible to realize 9 promising technologies among 13, and responses were investigated in order of “intelligent railroad maintenance system” (3 responses), “highly efficient hybrid railroad vehicle” (3 responses), “door-to-door advanced railroad system” (2 responses), “V2X autonomous cooperation road system” (2 responses), “autonomous driving railroad system” (2 responses), and “traffic infrastructure rapidly improving and replacing devices” (2 responses).¹

On the other hand, there were responses from the entire world as to how it was impossible to realize 2 promising technologies out of 13; “road user safety support facility” (1 response) and “traffic infrastructure rapidly improving and replacing devices” (1 response).

In addition, the proportion of 3 technologies among 13 turned out to be higher in “within 10 years” than “after 10 years” in Vietnam in terms of the period of realizing technology. In the world, the proportion of 8 technologies among 13 turned out to be higher in “within 10 years” than “after 10 years.” The response rate of “within 10 years” and “after 10 years” is the same for manned and unmanned autonomous aircraft designing, manufacturing, and certifying system².

■ Major research topics

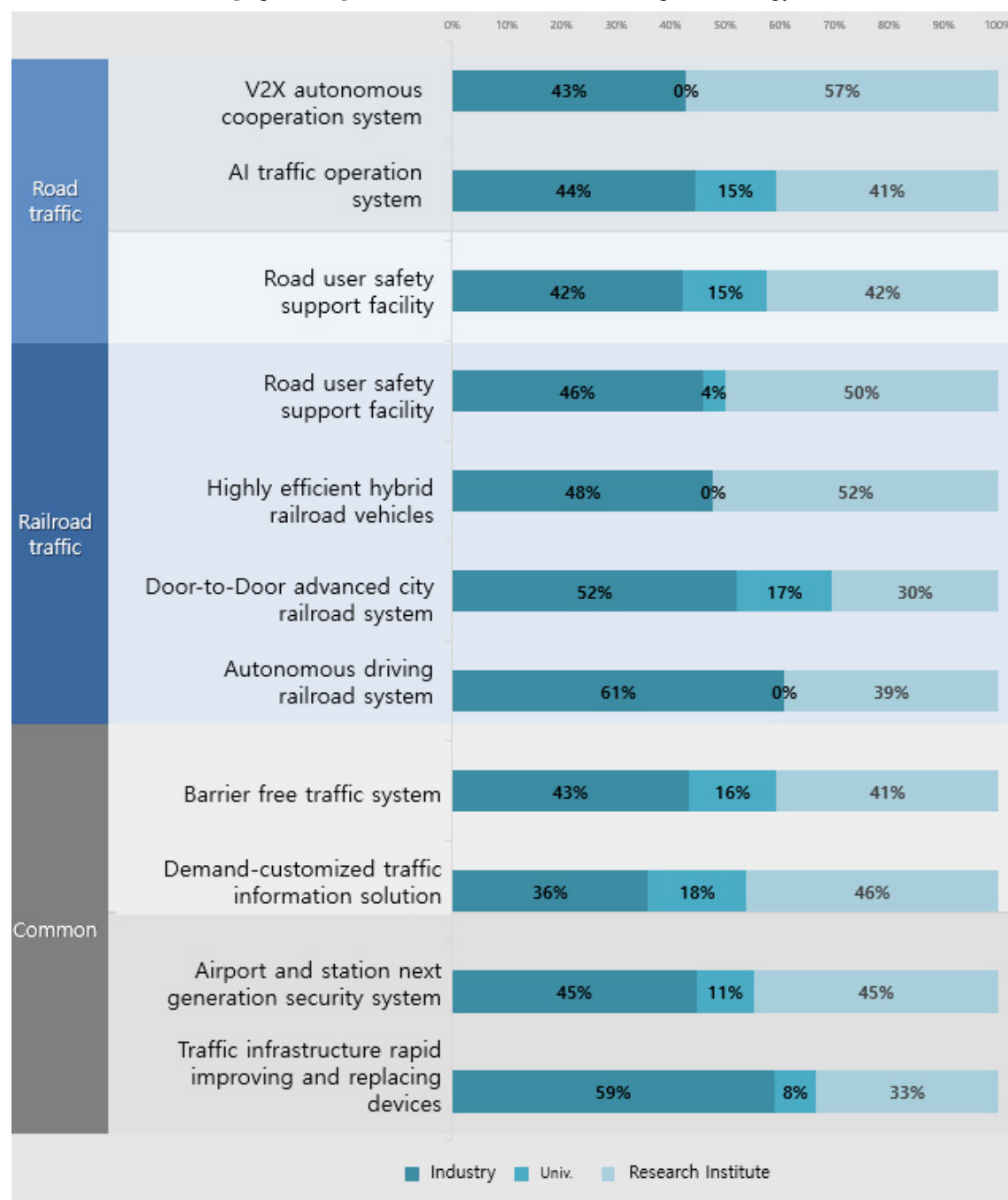
According to the results of researchers (academic and industrial collaboration) in Vietnam who were leading the research related to the realization of promising technologies in the future, there were 9 promising technologies with the highest response rate in “industry,” followed by 0 in “academe” and 6 in “research area.” The response rate in industry and research area is the same for “road user safety support facility” and “airport and station next-generation security system.”

² The response rate of “within 10 years” and “after 10 years” is the same for manned and unmanned autonomous aircraft designing, manufacturing, and certifying system.

[Table 3-7] Possibility of Realizing Each Promising Technology in the Future

Area	Classification		Within 10 years		After 10 years		Total	
			Case	Proportion	Case	Proportion	Case	Proportion
Road traffic	V2X autonomous cooperative system	Vietnam	11	42%	15	58%	26	100%
		World	22	79%	6	21%	28	100%
	AI traffic operation system	Vietnam	11	41%	16	59%	27	100%
		World	19	70%	8	30%	27	100%
	Road user safety support facility	Vietnam	16	64%	9	36%	25	100%
		World	20	80%	5	20%	25	100%
Subtotal		Vietnam	38	49%	40	51%	78	100%
		World	61	76%	19	24%	80	100%
Railroad traffic	Highly efficient hybrid railroad vehicle	Vietnam	7	33%	14	67%	21	100%
		World	15	63%	9	38%	24	100%
	Door-to-door advanced city railroad system	Vietnam	4	19%	17	81%	21	100%
		World	11	48%	12	52%	23	100%
	Autonomous driving railroad system	Vietnam	3	14%	18	86%	21	100%
		World	11	48%	12	52%	23	100%
	Intelligent railroad maintenance system	Vietnam	5	25%	15	75%	20	100%
		World	11	48%	12	52%	23	100%
Subtotal		Vietnam	19	23%	64	77%	83	100%
		world	48	52%	45	48%	93	100%
Aviation traffic	Autonomous aviation support infrastructure	Vietnam	1	33%	2	67%	3	100%
		World	1	33%	2	67%	3	100%
	Manned and unmanned autonomous aircraft	Vietnam	1	33%	2	67%	3	100%
		World	1	33%	2	67%	3	100%
Total		Vietnam	2	33%	4	67%	6	100%
		World	2	33%	4	67%	6	100%
Common	Barrier-free traffic system	Vietnam	14	38%	23	62%	37	100%
		World	23	62%	14	38%	37	100%
	Demand-customized traffic information solution	Vietnam	17	45%	21	55%	38	100%
		World	28	72%	11	28%	39	100%
	Airport and station next-generation security system	Vietnam	20	54%	17	46%	37	100%
		World	22	56%	17	44%	39	100%
	Traffic infrastructure rapidly improving and replacing devices	Vietnam	13	35%	24	65%	37	100%
		World	25	66%	13	34%	38	100%
Subtotal		Vietnam	64	43%	85	57%	149	100%
		World	98	64%	55	36%	153	100%
Total		Vietnam	123	39%	193	61%	316	100%
		World	209	63%	123	37%	332	100%

[Figure 3-5] Researchers to Lead Promising Technology



■ Policies required to realize technology

According to the investigation on public policies needed to realize promising technology in the future, “infrastructure construction” was perceived to be the most required policy to realize 9 technologies among 13, followed by “system improvement” and “cooperation promoting” policies in 3 technologies.

[Table 3-8] Policies Required for Each Promising Technology

Classification		1 st priority	2 nd priority
Road traffic	V2X autonomous cooperation road system	Infrastructure construction	Increase in R&D investment
	AI traffic operation system	Infrastructure construction	Increase in R&D investment, System improvement
	Road user safety support facility	Infrastructure construction	System improvement
Railroad traffic	Highly efficient hybrid railroad vehicle	Infrastructure construction, Increase in R&D investment	System improvement
	Door-to-door advanced city railroad system	Infrastructure construction	System improvement
	Autonomous driving railroad system	Infrastructure construction, System improvement	Increase in R&D investment
	Intelligent railroad maintenance system	Infrastructure construction	System improvement
Aviation traffic	Autonomous aviation support infrastructure	Cooperation promotion	Manpower cultivation, Infrastructure construction, Increase in R&D investment, System improvement
	Manned and unmanned autonomous aircraft designing, manufacturing, and certifying system	Infrastructure improvement, manpower cultivation	Cooperation promotion, System improvement
Common	Barrier-free traffic system	Infrastructure construction	Cooperation promotion
	Demand-customized traffic information solution	Cooperation promotion	Infrastructure construction
	Airport and station next-generation security system	System improvement	Cooperation promotion
	Traffic infrastructure rapidly improving and replacing devices	Infrastructure construction	System improvement

3. Guidelines for National Technology Foresight Exercise

These guidelines are expected to produce substantial references that can serve as a useful, practical guide to Vietnam in performing the National Technology Foresight Exercise successfully.

3.1 Scope

Vietnam's Technology Foresight Exercise will be carried out in 4 steps with the following scope:

Step 1 is intended to identify key stakeholders and build engagement. They should find and select the experts. They should also organize the committees and assign their roles and responsibilities.

Step 2 is intended to provide new perspectives about the future and identify key issues.

- Identify key issues that affect the future development of Vietnam by scanning the horizon of Vietnam's current and planned operating and peripheral environments.
- Study the background, future, and potential impacts of these issues.

Step 3 is intended to identify future technologies considered to be important in Vietnam to respond to key issues. A brief explanation of each future technology will be given by Vietnamese experts.

- Prepare an initial list of future technologies to fulfill key issues and needs.
- Search for previously identified future technologies as candidates from other foresights and S&T policies.
- Supplement the future technology list through experts' consultation.
- Adjust the depth and width of selected future technologies.

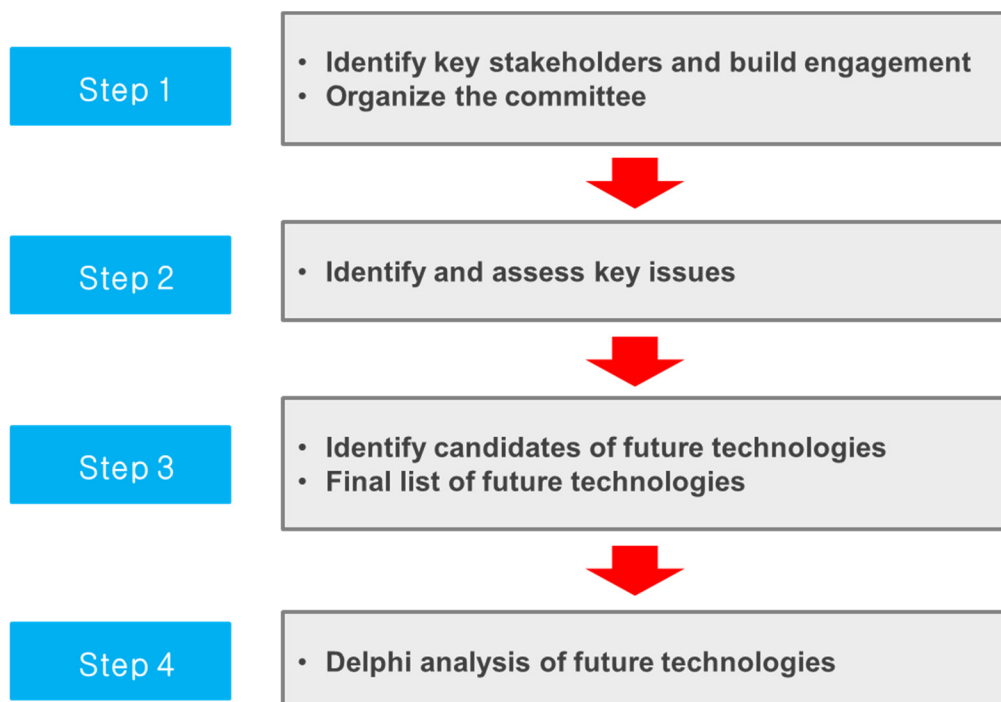
- Draw up the final list of future technologies.
- Provide a brief description of each future technology.

Step 4 is intended to analyze future technologies. A Delphi survey questionnaire will be prepared. Future technologies are analyzed in terms of their characteristics, importance, time of technology realization, and required government policies. It covers the following:

- Design the survey questionnaire.
- Run the survey.
- Analyze the results, report the outcome, and make recommendations.

3.2 Methods and Process

[Figure 3-6] Process of Technology Foresight Exercise



Step 1

Module 1: Identify key stakeholders and build engagement

- Stakeholders can be identified by considering the roles of scientist, government, non-government organization (NGO), industry, and citizen groups. It is important to recruit gifted individuals who are prepared to learn and share, not just present their organization's official positions. Such individuals can be located by a search through databases and web resources or by seeking advice from other informed people.
- It is also important to build wide and in-depth engagement between stakeholders.

Module 2: Organize the committees

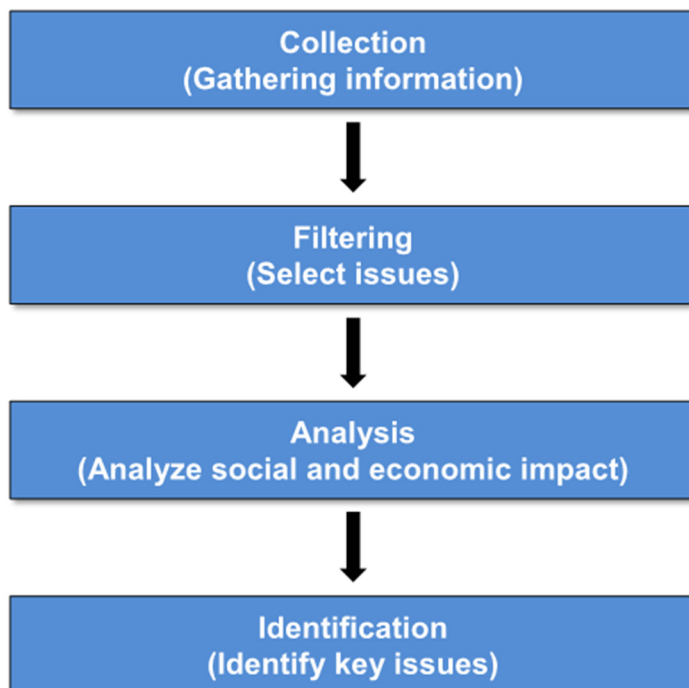
- Tasks and responsibilities of working groups, panels, committees, and sponsoring agencies should be assigned. A steering committee and a management team are generally established. Expert panels are also created to focus on particular issues.

Step 2

Module 3: Identify and assess key issues

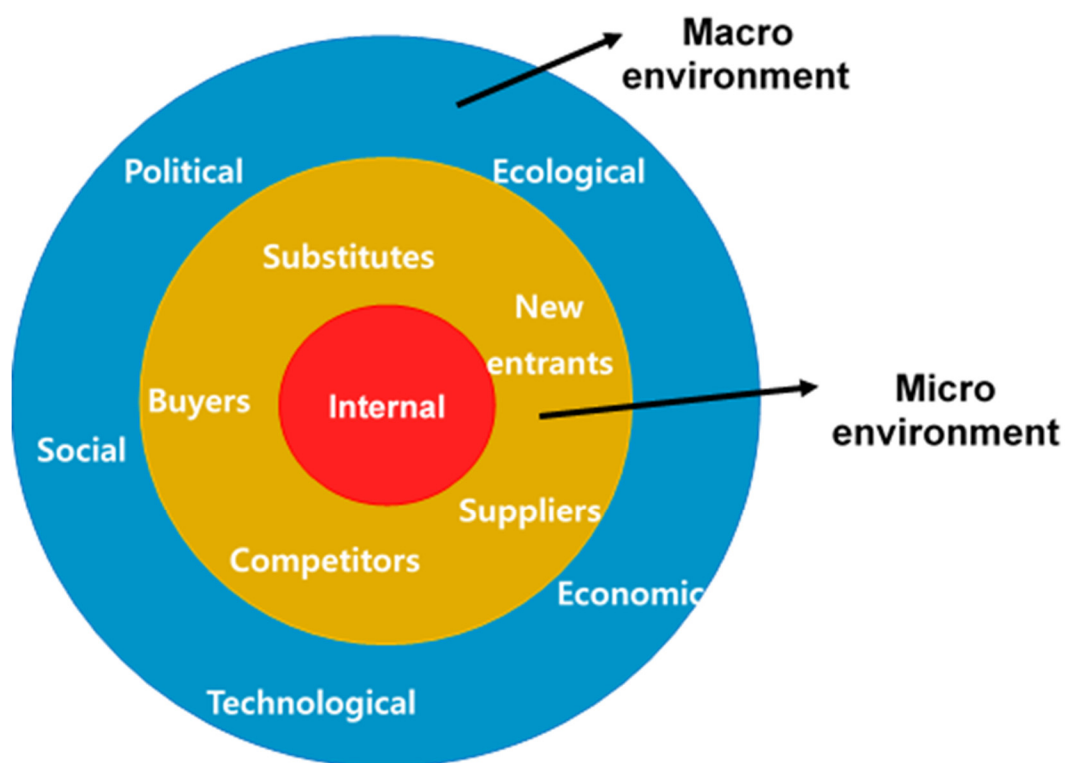
- Collection: Potential future trends/issues are gathered from literature reviews of books, journals, technological reports, trend studies and magazines, and websites and by consultation or survey with expert panels and stakeholder.
- Filtering: Issues are identified/selected from potential future issues by STEEP analysis, which can gauge how the external environment will impact the future development of Vietnam. The STEEP factors include external macro-environmental factors and Vietnam's special issues.

[Figure 3-7] Process for Identifying Key Issues



- Collection: Potential future trends/issues are gathered from literature reviews of books, journals, technological reports, trend studies and magazines, and websites and through consultation or survey with expert panels and stakeholder.
- Filtering: Issues are identified/selected from potential future issues by STEEP analysis, which can gauge how the external environment will impact the future development of Vietnam. The STEEP factors include external macro-environmental factors and Vietnam's special issues.

[Figure 3-8] STEEP Analysis



- Analysis: These issues are further analyzed in terms of social and economic impact on Vietnam. The opinion of stakeholders on these issues will also be evaluated.
- Identification: Key issues are identified. An analysis report of these key issues will be provided.

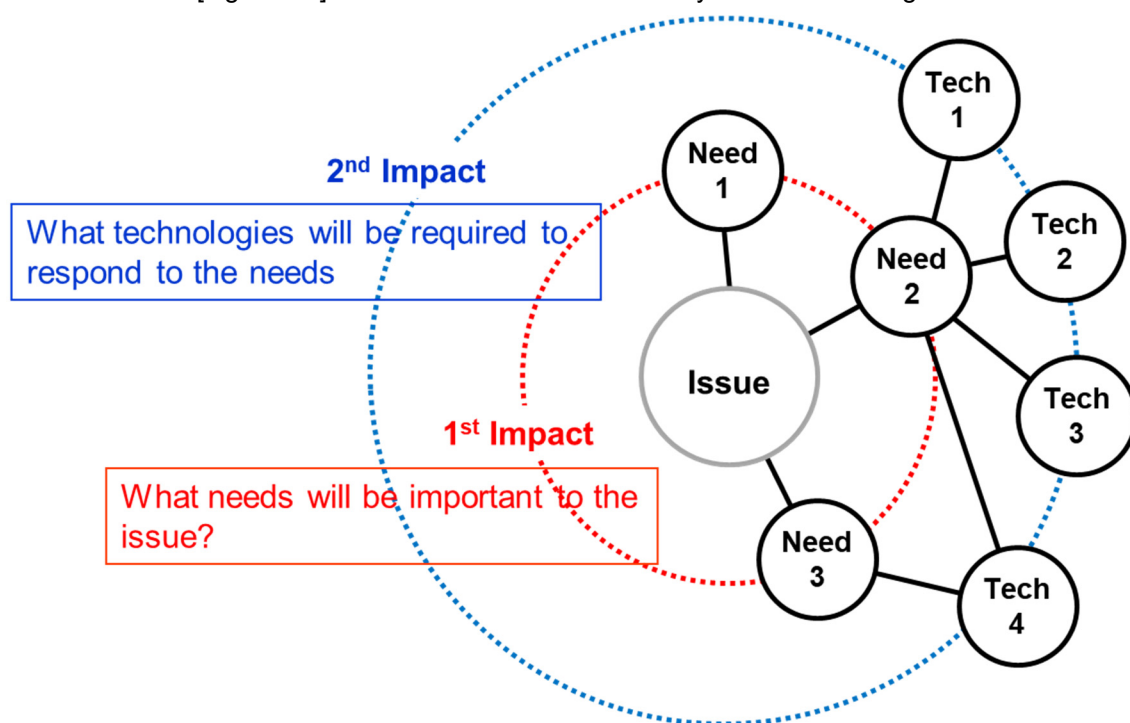
Step 3

Module 4: Identify candidates of future technologies

- Future technologies considered to be important in Vietnam to respond to key issues will be identified through literature search, brainstorming of experts in the workshop, and/or circumstances scanning. The depth and width of initial future technologies in the list should be adjusted.

- The Futures Wheel method can be used to identify future technologies in the expert workshop. Futures Wheel is a method for the graphical visualization of direct and indirect future consequences of a particular change or development. It is a simple technique of exploring the future, requiring only blank paper, pen, and one or more fertile minds.

[Figure 3-9] Futures Wheel Method to Identify Future Technologies



Module 5: Draw up the final list of future technologies

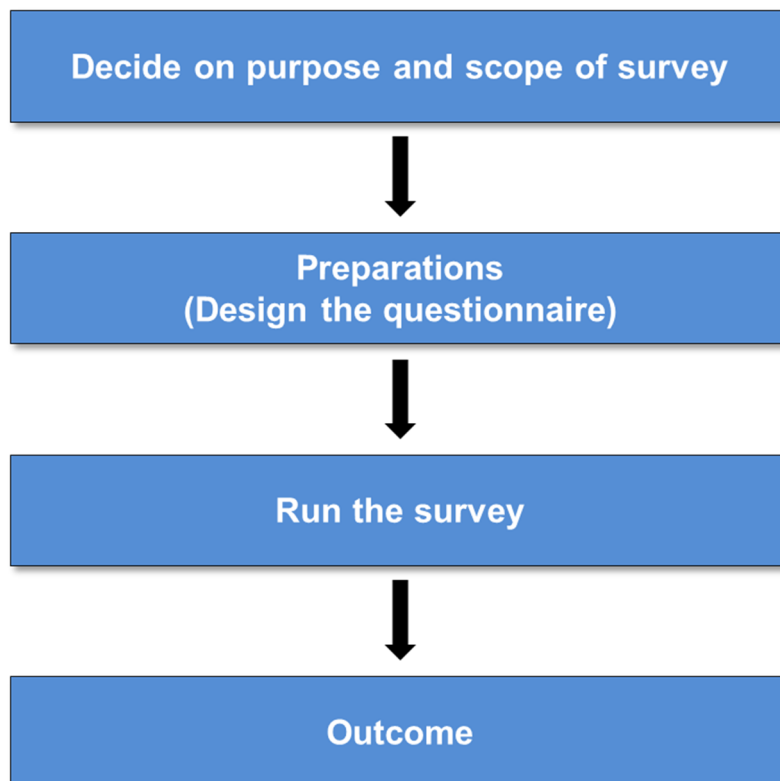
- In this module, expert panel discussion and/or industrial interview will be used to rescreen the first identified pool. Experts in the 1st screening can participate in this step, but setting a limit by high-level experts is also possible.
- Once the final list of future technologies is identified, a brief description will be attached to each future technology, specifying its main characteristics, application areas, and critical problems to be addressed.

Step 4

Module 6: Conduct a Delphi survey of future technologies

- Delphi is a well-established expert-based technique that recognizes human judgement as legitimate and useful input for generating forecasts and exploring the future impacts of science and technology.
- Surveys are normally carried out in rounds or waves (normally two, rarely three). The second round will be conducted in an almost similar way to the first round. Since median values and/or share of each technology will be offered to the respondents to change their first answer, however, consensus between respondents will be induced.

[Figure 3-10] Delphi Process



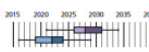
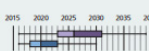
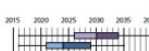
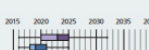
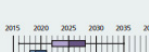
- Questions in the survey should be clearly and precisely phrased, and parameters used in the survey should be comprehensively explained. This is to ensure minimal difference in the understanding of survey participants, who come from various fields and backgrounds (i.e., Science and Technology and Socio-economics). It has some common variables:
 - degree of expertise in or level of knowledge of each topic
 - time of occurrence
 - impact assessment
 - constraints / limitations
 - necessity of cooperation
 - level of competitiveness
 - desirability and likelihood of the topics
 - importance for the country

Module 7: Analyze the Delphi results and make recommendations

- From the Delphi survey results, future technologies are analyzed to assess their characteristics, importance, time of technology realization, and required government policies.

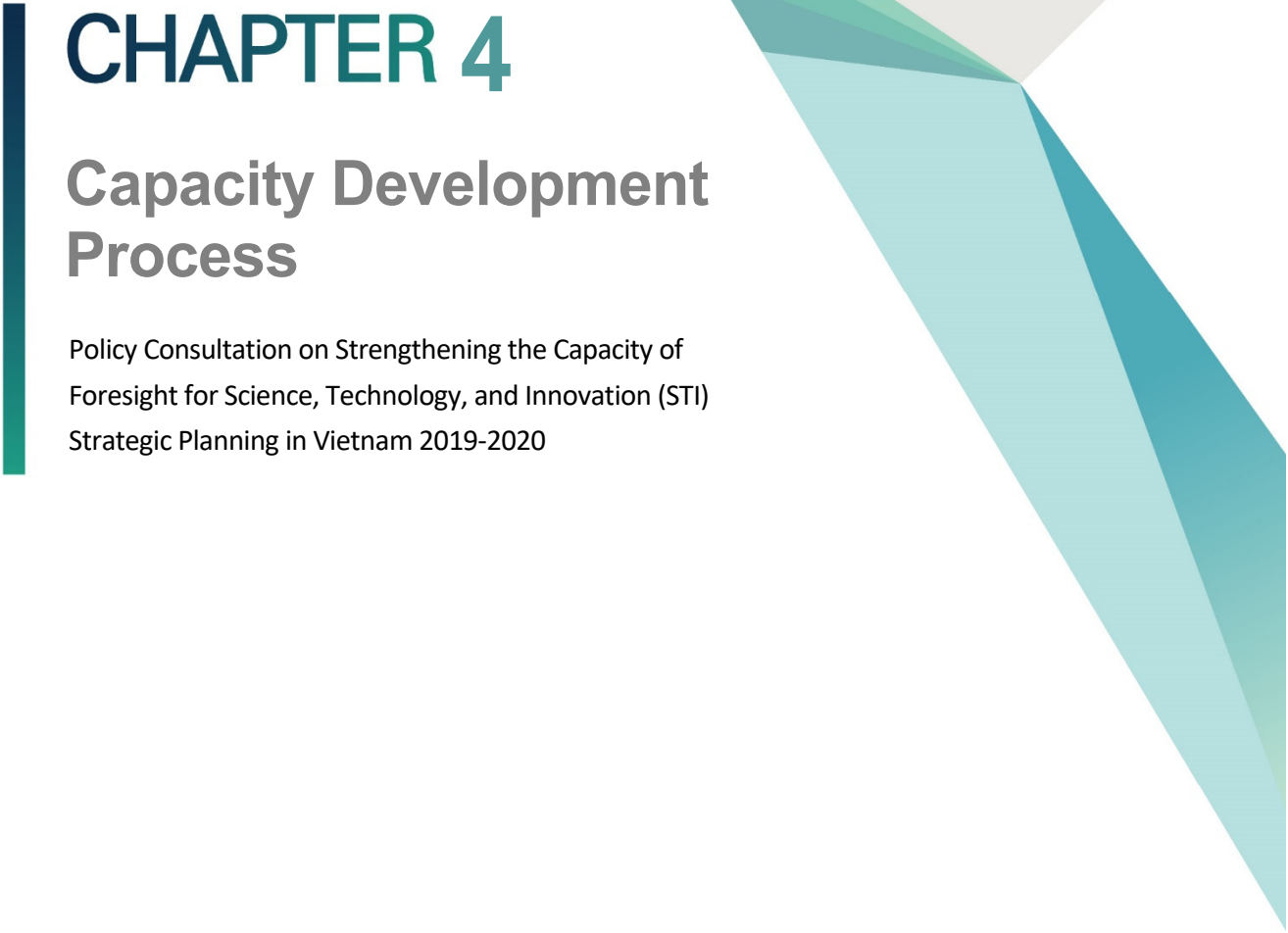
Chapter 3. Sharing Korea's Experiences in Science and Technology Planning and the Technology Foresight

[Figure 3-11] Results of the Delphi Survey Analysis of Future Technologies in the 5th Science and Technology Foresight

No.	Future Technology	Technological Criteria (5-point System)			Significance (5-point System)			Year Realized (Year)			Government Policy (Top Priority)
		Innovativeness	Uncertainty	Negative Impacts	Scientific	Public	Economic Industrial				
1	Integrated High Altitude Wide Area Environment and Facility Monitoring SystemIntegrated High Altitude Wide Area Environment and Facility Monitoring System	4.3	3.5	2.4	4.2	4.7	4.0	Domestic	2028		Increase in Research Fund
								Overseas	2022		
2	Real-time Continuous Disaster Monitoring Technology using Remote Sensing Information of Multiple Satellites	4.0	3.1	2.4	4.3	4.6	4.1	Domestic	2026		Increase in Research Fund
								Overseas	2020		
3	Integrated Global Issue (Environment, epidemic, disaster, etc.) Interconnection and Analysis Modelling and Simulation Software	4.3	3.6	2.7	4.3	4.4	3.9	Domestic	2030		Infrastructure Creation
								Overseas	2024		
4	Automated Disaster Information Collection • Transmission Technology Using UAV and Drones	4.2	2.8	2.5	4.3	4.5	4.2	Domestic	2023		Institutional Reform
								Overseas	2019		
5	Observation-based Integrated Earthquake Early Warning System based on Facility Management Conditions	4.0	3.4	2.3	4.3	4.8	4.3	Domestic	2024		Infrastructure Creation
								Overseas	2019		
6	Decision-making Support Technology for Emergency Evacuation and Rehabilitation	4.0	3.2	2.3	4.2	4.8	4.3	Domestic	2025		Increase in Research Fund
								Overseas	2019		

- Prepare the final report and incorporate participants' feedback and comments in your reports.

- Global convergence trends over the past few decades (3As: adopt, adapt, advance).
- When the national innovation system has more functional components in it, the effectiveness of TF projects would increase.
- Having consultation from a wider range of experiences, involving many aspects of domain beyond one specialized features, which would also engage broader sectors and stakeholders.

A decorative vertical bar in dark teal is positioned to the left of the chapter title. To the right, there are abstract geometric shapes: a large light grey triangle pointing downwards, and several overlapping triangles in various shades of teal and light blue, also pointing downwards.

CHAPTER 4

Capacity Development Process

Policy Consultation on Strengthening the Capacity of
Foresight for Science, Technology, and Innovation (STI)
Strategic Planning in Vietnam 2019-2020

Chapter 4. Capacity Development Process

1. Kick-off Meeting and Field Visit

1.1 Introduction (objectives, team, schedule, etc.)

1.1.1 Objectives

The purpose of the first visit is to have an inception meeting and a workshop for knowledge sharing. The inception meeting shall involve project counterparts, VISTI, and STEPI to discuss the project purpose, activities, and expected outputs. The workshop, titled “National Future Strategy and Foresight,” will be a public event for sharing knowledge on the national STI strategic foresight plans or activities of Vietnam and Korea. Various stakeholders including government officials from the relevant ministries and directors of the research institutions will be invited to the event to build consensus on the importance of strengthening the national strategic planning system and the directions for foresight capacity building. This event will identify policy demands for the main institutions and ministries (Jungwon Lee, et al, 2019).

The following are the three main goals of the visit:

- Confirm the project purpose and expected outcome.
- Collect the necessary information for assessing Vietnam’s national STI strategic planning and technology foresight capacities.
- Share Korean experiences in national strategic planning and technology foresight.
- Exchange ideas and opinions on capacity-building programs.

The main activities of the workshop will be as follows:

- Short remarks from the STEPI project leader on the project overview including the purposes, scope, methodology, implementation, planned schedule, and results
- Presentations by the Korean experts on Korean cases of national strategic planning and technology foresight
- Presentations by the Vietnamese experts on Vietnam's STI policy and strategy, the development and application of technology foresight, and a few cases, to be followed by a lively discussion on key issues and the exchange of ideas and opinions
- Follow-up in-depth meetings with stakeholders to collect their personal views and insights on improving foresight and strategic planning capacities

1.1.2 Team

The research team was responsible for holding the kick-off meeting and field visit to Vietnam. They conducted interviews with the stakeholders from VISTI and other research institutions and decision makers including the Deputy Minister of Science and Technology of Vietnam (Jungwon Lee, et al, 2019).

[Table 4-1] Kick-off Meeting and Field Visit Research Team

Organization	Name	Project role/Position	Contact
STEPI	Dr. Jungwon Lee	Project Leader Senior Research Fellow	leejw@stepi.re.kr
	Dr. Byeongwon Park	Foresight Expert Research Fellow	bwonpark@stepi.re.kr
	Ms. Yongin Choi	Project Coordinator Researcher	luciachoi@stepi.re.kr
Ewha University	Prof. Young Il Park	Adviser / Speaker Professor	yipark@ewha.ac.kr
KIST	Dr. Yoohyung Won	Adviser / Speaker Principal Researcher	yhwon@kist.re.kr

1.1.3 Schedule

[Table 4-2] Kick-off Meeting and Field Visit Schedule

Date/Time		Institution	Description
Start	End		Activities and Time
DAY 1: Monday, April 15, 2019			
		Arrival	ICN-HAN (Jungwon Lee, Yong-in Choi)) 21:50 by KE0679 FRA-HAN (Byeongwon Park) 05:45+1 by VN0036
DAY 2: Tuesday, April 16, 2019			
09:00	12:00	Hotel	Internal Meeting – STEPI team
12:00	14:00		Lunch
14:00	17:00	Korean Embassy to Vietnam	Interview with the Ambassador and Counselor • Introduction of the project • Discussion on Vietnam’s STI policy issues and expected outcomes • Discussion on the <u>agenda</u> for further cooperation on the project
DAY 3: Wednesday, April 17, 2019			
09:30	12:00	KOICA	Interview with the Representative • Introduction of the project • Discussion on Vietnam’s STI policy issues and relevant ODA projects • Discussion on the <u>agenda</u> for further cooperation on the project
12:00	14:00		Lunch
14:00	17:00	<u>VISTI</u>	Inception Meeting • Introduction of the project • Discussion on the project purpose, <u>main</u> activities, schedule, and expected outcome • Exchange of ideas and views on the project details • Identification of policy demands for the project
		Arrival	ICN-HAN (Yoohyung Won) 13:05 by KE0483 ICH-HAN (Young Il Park) 21:50 by KE0679

Date/Time		Institution	Description
Start	End		Activities and Time
DAY 4: Thursday, April 18, 2019			
09:30	12:30	Army GuestHouse Hotel	Workshop on National Future Strategy and Technology Foresight
12:30	14:00		Lunch
14:00	17:00	VISTI	In-depth Discussion • Discussion on the results of the workshop • Interview with various internal focus groups • Identification of key challenges in policy and in the project • Exchange of views and insights on the current status and capacity-building programs for Technology Foresight and Strategic Planning
DAY 5: Friday, April 19, 2019			
11:00	16:00	VKIST	Interview with the Representative • Introduction of the project • Discussion on Vietnam’s STI policy issues and relevant ODA projects • Discussion on the agenda for further cooperation on the project
		Departure	HAN-ICN (Jungwon Lee, Yong-in Choi, Young Il Park, Yoohyung Won) 23:10 by KE0680 HAN-FRA (Byeongwon Park) 22:55 by VN0037

1.2 Kick-off Meeting

1.2.1 Kick-off Meeting and Field Visit Participants' List

[Table 4-3] Kick-off Meeting and Field Visit Participants' List

Name	Organization
Kim Jin Oh	KOICA Vietnam Office / Chief Officer
Lim In Young	KOICA Vietnam Office / Deputy Chief Officer
Kim Dong Bae	Korean Embassy to Vietnam / Counselor
Shin Dong Chan	Korean Embassy to Vietnam / Deputy Director
Shin Sang Yeol	Korean Embassy to Vietnam / Director
Ma. Nguyen Hoang Hai	VISTI / Director
Bach Tan Sinh	VISTI / Director
Kum Dong Hwa	VKIST / President
Hoang Van Cuong	Central Institute for Economic Management
To Mai Trinh	Department of International Cooperation
Yang Hi Sang	Hanoi University of S&T
Phan Van Nghia	Ministry of Information and Communication
Le Dinh Tien	Ministry of Science and Technology
Luu Quang Minh	Ministry of Science and Technology
Giang Manh Khoi	National Center for Technological Progress
Tran Hong Minh	National Center for Technological Progress
Nguyen Viet Hoa	National Institute of S&T Policy and Strategy Studies
Ta Doan Trinh	National Institute of S&T Policy and Strategy Studies
Duong Khanh Dung	National Institute of Information and Communications Strategy
Hoang Anh	National Office of Intellectual Property of Vietnam
Tran Khắc Trọng Tài	National Office of Intellectual Property of Vietnam
Pham Quang Tri	National Technology Innovation Fund
Ngo Ha	Science and Development Newspaper
Le Trong Tai	State Agency for Technology Innovation
Nguyen Duc Hoang	State Agency for Technology Innovation
Le Trong Nghi	TDST

Name	Organization
Nguyen Hanh	Vietnam Center for Science and Technology Communication
Ngo Thi Loan	Vietnam Center for Science and Technology Evaluation
Pham Hong Bach	Vietnam Center for Science and Technology Evaluation
Doan Thi Thu Ha	Vietnam Institute for Development Strategies
Nguyen Thi Nguyet	Vietnam Institute for Development Strategies
Vu Thi Hai	Vietnam Intellectual Property Research Institute
Vu Van Hung	Vietnam Journal of Science, Technology, and Engineering
Le Ngoc My	VISTI
Nguyen Van Hoa	VISTI
Nguyen Van Khuong	VISTI
Nguyen Viet Hung	VISTI
Tran Van Tam	VISTI
Le Vu Toan	VISTI

1.2.2 Kick-off Meeting's Main Activities and Agendas

At the kick-off meeting with VISTI, the objectives and scope of the Project were discussed. Further specific action plans were discussed, and STEPI and VISTI agreed on the main items of the project programs and activities. Since the kick-off meeting was held after the workshop on national future strategy and technology foresight, the researchers of both counterparts shared insights on Korea's S&T strategic planning and its implications on Vietnam's future policy direction. The key points of the meeting can be summarized as follows:

- Agreed on the Project Action Plan
- Discussed VISTI and other institutions' role in Vietnam's STI system
- Shared Vietnam's challenges in technology foresight and national STI strategies
- Shared Korean policy experiences in national STI strategies based on technology foresight

1.3 Field Visit (information by organization to visit)

1.3.1 Meeting with the Embassy of the Republic of Korea in Hanoi, Vietnam

A meeting with the Korean Embassy in Vietnam was held on April 16, 2019 to provide an introduction of the Project. The Korean Embassy shared the current status of Vietnam's STI policies and the STI-related ODA program. The key results of the meeting are as follows (Jungwon Lee, et al, 2019):

- Current status of Vietnam's STI policy and ODA programs
 - The main challenges of Vietnam's STI policy are "unclear direction and insufficient contents," "STI-related data system," and "insufficient R&D evaluation system."
 - The main challenges of Vietnam's ODA programs (as a recipient country) are "sustainability" and "redundancy of programs."
- Issues to be considered to maximize the effectiveness of the Project
 - Need for cooperation by Korean R&D companies located in Vietnam
 - Cooperation opportunities with V-KIST and other ODA projects of Korea for Vietnam

1.3.2 Meeting with KOICA Vietnam Office

The research team had a meeting with the KOICA Vietnam office. The basic information of the Project were shared, and additional issues on Vietnam's policies were discussed.

- Possibility of cooperation with KOICA
 - Prioritize identification of cooperation opportunities with V-KIST.
 - Discuss possible ways of cooperative activities considering the role of V-KIST in Vietnam's national STI system.
 - Recently, KOICA has not conducted STI-related ODA projects for Vietnam as there is little demand from the Vietnamese government in the STI field.

- Issues to be considered
 - The capacity of Vietnam's human resources is much more competitive compared to that of other South Asian countries.
 - For the effective progress of the Project, the local experts need to be involved actively.
 - Considering Vietnam's administrative customs and culture, clear agreement on the Project budget allocation and cooperative response to document-based process are needed.

1.3.3 Meeting with Vietnam's Deputy Minister of Science and Technology and VKIST

The research team had a meeting with Vietnam's Deputy Minister of Science and Technology and the President of VKIST. The in-depth discussion highlighted the following key points:

- Current status of Vietnam's technology foresight and S&T capacity
 - The Vietnamese government is trying to enhance the effectiveness of implementing the national S&T policies by drawing up quantitative goals and specific action plans in the National Plan 2021-2030.
 - Vietnam needs to strengthen the science and technology policy capacity of Vietnamese policymakers.
 - Vietnamese research institutions need to be systemized with roles that are distinct from each other.
- Several ways to enhance the effectiveness of the Project
 - SATI and VISTI can assume a role in the Project, since each has data and human resources related to technology foresight.
 - The Project could link its activities with other small and large national projects that involve technology foresight and innovation policies.

- Expected outcome of the Project
 - The Project, which aims to enhance Vietnam's capacity for technology foresight, can be the steppingstone for further cooperation activities between Korea and Vietnam on S&T policies.
 - The Project can be linked with VKIST's planned role expansion in analyzing Vietnam's economic and technological capacities.

1.4 Workshop on National Future Strategy and Technology Foresight

1.4.1 Concept Note

a) Background

Technology development is much faster and broader than many people realize. It affects every aspect of human life. Digital technology revolution such as artificial intelligence, big data, Internet of Things, block chain technology, 5G, etc. even opens up new possibilities in triggering industrial transformation. Such may result in shifting the global order or solidifying the current one. It is unclear at the moment who will be the next winner.

The future is shaped and influenced by the decisions made today. Still, making the right decision at the right time to exploit the full potential of new technology and innovation becomes more difficult, probably almost impossible. The time allowed for decision making is getting shorter in contrast to the increasing degree of complexity and uncertainty of issues. Though the volume of data is getting huge, insight by sense making is too shallow. The only way to adapt and survive in the future is to embrace the complexity and uncertainty as the inherent characteristics of the system instead of trying to predict the future. It is impossible to predict the future by forecasting through trend extrapolations.

b) Foresight capacity to shape the future

Foresight refers to approaches to informing decisions made robustly by improving inputs concerning the longer-term future and by engaging a wider range of social stakeholders and linking the strategic vision to present-day actions. Foresight will enhance the capacity for decision making and promote public debates by providing opportunities to discuss longer-

term trends as well as the expected outcome or implications of short-term decisions. Likewise, foresight provides alternative interactive learning platforms for robust decision making in the context of uncertainties associated with the 4th industrial revolution.

c) Goal of the project

Nonetheless, realizing the importance of long-term thinking and the need for running a foresight exercise is different from the action on vision from the foresight results. In many cases, foresight has limited influence on changing the behavior and action of the decision makers due to many causes such as unclear goals, governance, incentives and cultures, budgets, etc. (Jungwon Lee, et al, 2019).

Since the lack of competence among decision makers turns out to be one of the main reasons, the primary goal of the current project is to enhance the foresight capacity of the decision makers through the pilot-scale foresight exercise, focusing mainly on research and innovation-driven future development. The opportunities provided by such innovations intimately shape most areas of human social life (Jungwon Lee, et al, 2019).

Structured approaches are employed to build capacities in foresight, engagement, and integration among the decision makers. The approaches will be interactive and participative, focusing on long-term social, economic, and technological developments relevant to the country. Based on the project's goal, this workshop is designed to be the first step of the project for sharing the current status of Vietnam's national S&T policy and Korea's experiences in strategic planning and technology foresight. All the participants are invited to discuss the agenda and its implications on Vietnam's national STI strategy (Jungwon Lee, et al, 2019).

1.4.2 Program

Date: Thursday, April 18, 2019

Time: 9:00am – 12:40pm

Venue: Army Hotel, 33 C Pham Ngu Lao, Hanoi, Vietnam

Expected Participants: STEPI, VISTI, VKIST, government officials, other relevant experts

[Table 4-4] Program of Workshop on National Future Strategy and Technology Foresight

Time	Activity	Speakers
8:40 - 9:00	Registration	
9:00 - 9:15	Opening Session	
9:00 - 9:05	Welcome Remarks Introduction of the Project	Dr. Jungwon Lee Senior Research Fellow, STEPI
9:05 - 9:10	Welcome Remarks	Dr. Dongwha Kum President of VKIST
9:10 - 9:15	Opening Remarks	Dr. Nguyen Hoang Hai , Acting Director, Department of Science Administrator and Training, VISTI
9:15 - 12:30	Presentation Session	
9:15- 9:55	Presentation 1 Introduction to National Strategic Planning and Korean Case of STI Development • Overview of 4 Korean Cases of Long-term STI Development Plan • Basic Framework for the STI Development Plan • Key Issues, Opportunities, and Challenges • Lessons Learned and Implications for the Next Steps	Prof. Young Il Park Professor, Ewha University Former Vice-minister, MOST
9:55- 10:25	Presentation 2 Introduction of Technology Foresight and Its Implementation • Understanding of Social Change • Reasons for the Recent Boom of Foresight • Procedure for Foresight • Linking Foresight to Policymaking • Technology Foresight to Wiring NIS * In this presentation, the success factors and barriers of the foresight exercise from the practitioner's perspective will be given to stimulate further discussion.	Dr. Byeongwon Park Research Fellow, STEPI

Time	Activity	Speakers
10:25 - 10:55	Presentation 3 How Korean GRIs are Preparing for the Future • KIST's Strategy for Redefining R&R • KIST's Alignment with the National STI Strategic Goals • KIST's Contributions to Korean STI Policy: K-DARPA Program	Dr. Yoohyung Won Principal Researcher, KIST
10:55 - 11:10	<i>Coffee Break</i>	
11:10 - 11:30	Presentation 4 Overview of Vietnam's STI Policy Strategy • Some General Context for STI Policy and Strategy Process for Vietnam • Case of Using Foresight for STI Strategy 2020: Some Experiences • Potential Implications	Dr. Tran Ngoc Ca Former Vice-President of NISTPASS
11:30 - 11:50	Presentation 5 Development and Application of Technology Foresight in Vietnam • Overview of the Current Stage of Developing and Applying Technology Foresight in a Few Sectors • Lessons Learned for Making this Tool More Effective in the Context of Transforming the Current Strategic Mid-term and Long-term Planning of Science and Technology into the Engagement of a Wider Range of Stakeholders	Dr. Bach Tan Sinh Senior Research Fellow, VISTI
11:50 - 12:10	Presentation 6 Vietnam's Technology Foresight Case - Vietnam's Future Digital Economy Project • Approach Applied • Recent Development of Vietnam During the Course of the 4th Industrial Revolution and Digital Transformation • Some Major Trends and Formulation of Development Scenarios • Some Results from Modeling and Recommendations	Dr. Nguyen Duc Hoang Senior Officer, State Agency for Technology Innovation (SATI)/MOST
12:10 - 12:40	Floor Discussion	All Participants
12:40 - 12:45	Closing Remarks	
	Lunch	Army Hotel

1.5 Follow-up Activity (Meeting Recap)

The follow-up document was prepared in April 2019 and signed by STEPI and VISTI. The key actions required for the next steps are included. The main contents of the meeting recap are as follows:

a) Roles and responsibilities of STEPI and VISTI

- The roles and responsibilities of STEPI and VISTI shall follow those mentioned in PCP, in principle.

b) Major activities for 2019 and actions required

Kick-off meeting and first workshop

- The Kick-off Meeting and Workshop on National Future Strategy and Technology Foresight were conducted on April 15-19, 2019 in Hanoi, Vietnam.
- The summary of the major issues will be shared by STEPI and VISTI.

Capacity-building workshop in Korea

- The Capacity-building Workshop will be held in late August.

➔ Actions required

- By mid-June, STEPI will share the draft of the program with the official letter and other technical issues (including information on the venue, number of invitees, and budget support)
- By the end of July, VISTI will confirm the list of participants and proceed with the necessary administrative process.

Draft implementation plan for the pilot technology foresight of Vietnam

- During the project period, STEPI will guide and review the plan for pilot study.
- VISTI shall select the specific sector/area/challenge that requires technology foresight; this can be figured out throughout the seminar or workshop.

➔ Actions required

By August and during the capacity-building workshop, VISTI shall suggest one or two national goals as candidate topics for technology foresight. During the period July-August, STEPI and VISTI shall confirm the direction of the pilot study.

Final workshop

- To be discussed later

c) Other issues

Composition of the Project Team

- The internal team members of STEPI and VISTI have been selected.
- STEPI and VISTI will invite some external experts who will be involved in the project.
- The honorarium for the advisory activities of the external experts will be paid by STEPI based on its administrative criteria.

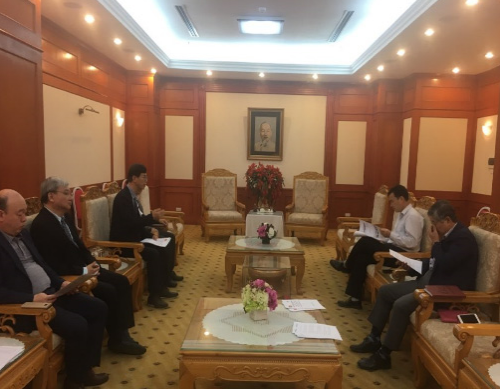
➔ *Actions required*

STEPI and VISTI shall share the list of external experts in Vietnam who are recommended by VISTI (and STEPI if applicable) ASAP.

Information to be provided by VISTI

- VISTI's mandate (already delivered)
- Organizational structure of Vietnam's S&T Governance including important institutions
- To be added during the Project period

[Figure 4-1] Kick-off Meeting, Seminar, and Field Study

Visit to the Korean Embassy in Hanoi, Vietnam	Visit to the KOICA Office in Vietnam
	
Kick-off Meeting	Workshop 1
	
Workshop 2	Visit to VKIST & MOST
	

2. Capacity Development Workshop

2.1 Introduction

2.1.1 Objective

The workshop aims to enhance Vietnam's capacity in Science, Technology, and Innovation (STI), especially focusing on technology foresight. The senior-level officials and managers of VISTI, relevant ministries, academe, and research institutions will be invited to participate in training programs. The lectures consist of various training courses from the introduction to technology foresight to hands-on project exercises (Jungwon Lee, et al, 2019).

2.1.2 Process

The workshop was supposed to be held in late August but has been postponed due to Vietnam's legal process (Jungwon Lee, et al, 2019).

During the period July to early August, STEPI and VISTI prepared for the workshop program and invitation of Vietnamese experts (Jungwon Lee, et al, 2019).

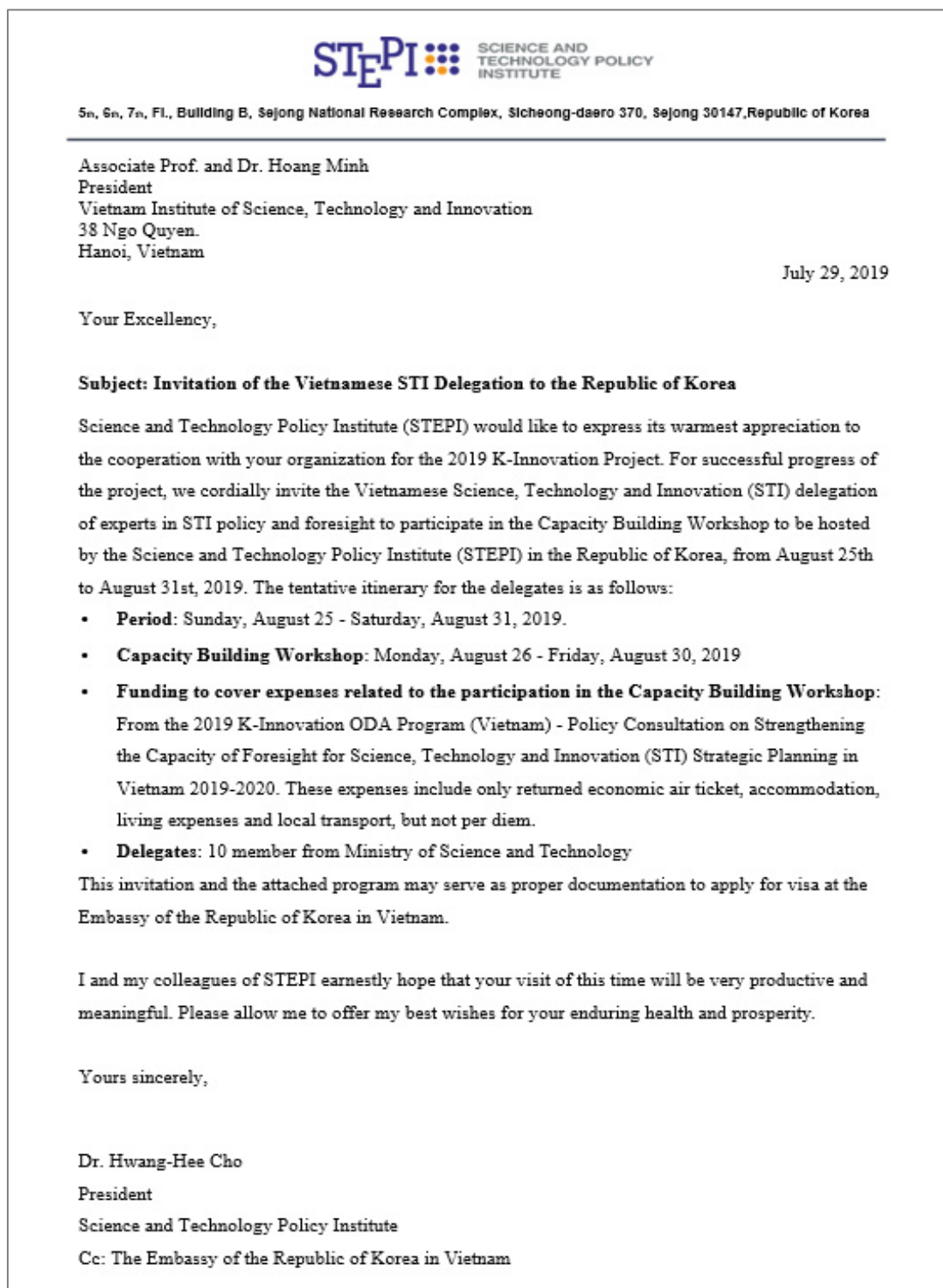
The following shows the program draft and the invitation letter sent by STEPI to VISTI:

[Table 4-5] Workshop Program (draft)

Date/Time		Venue	Description
Start	End		Activities and Time
DAY 1: Sunday, August 25, 2019			
		-	Arrival: HAN-ICN
DAY 2: Monday, August 26, 2019			
15:00	18:00		Orientation
18:00	19:00		Dinner
DAY 3: Tuesday, August 27, 2019			
09:00	12:00		Lecture 1 Introduction to the Korean STI System

Date/Time		Venue	Description
Start	End		Activities and Time
12:00	14:00		Lunch & Break
14:00	18:00		Lecture 2 Future Study and STI Strategy
18:00	19:00		Dinner
DAY 4: Wednesday, August 28, 2019			
09:00	11:00		Lecture 3 Introduction to Technology Foresight Korea's Experience: National Technology Foresight
11:00	13:00		Lecture 4 Introduction to the Technology Foresight Pilot Project (hands-on project) Step 1: Environment Analysis (Value Chain Analysis & SWOT Analysis)
13:00	14:00		Lunch
14:00	18:00		Lecture 5 Step 2: Identify the List of Products with Priority
18:00	19:00		Dinner
DAY 5: Thursday, August 29, 2019			
09:00	13:00		Lecture 6 Step 3: Identify the List of Future Technologies Step 4: Analyze Future Technologies with the Delphi Technique
13:00	14:00		Lunch
14:00	18:00		Lecture 7 Step 4: Analyze Future Technologies with the Delphi Technique (continued) Wrap-up
18:00	19:00		Dinner
DAY 6: Friday, August 30, 2019			
09:00	18:00		Site Visit - KIST or KISTEP - Gwacheon National Science Museum
DAY 7: Saturday, August 31, 2019			
		-	Departure: ICN-HAN

[Figure 4-2] Invitation Letter from STEPI to VISTI



2.1.3 Workshop Lecture Material

[Figure 4-3] Technology Foresight Manual

1. Introduction

1.1. Background

- There are significant uncertainties about the future shape of markets, governance and social values that will have an important impact on organizations and their capacity to meet their objectives
- The rapid technological and social change and uncertainties make the upsurge of interest in technology foresight, giving rise to its emergence as a global concept and policy tool
- Technology foresight can make unique contributions in the S&T policy to set research priority, to build networks, and to build visions of the future

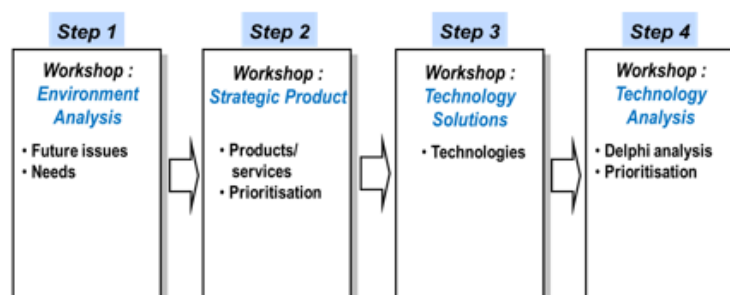
1.2. Purpose

- The purpose of this study is to provide the basic concepts of technology foresight and our experiences.
- This technology foresight study is expected to produce substantial references that can serve as a useful and practical guide to Vietnam when it plans to perform its national technology foresight exercise

1.3. Scope of the work

- This technology foresight study will be carried out in 4 work steps with the following scope:

- ✓ *Step 1 is intended to provide new perspectives about future, and identify issues and trends*
- ✓ *Step 2 is intended to identify STRATEGIC PRODUCTS which are considered to be important in the future of Vietnam*
- ✓ *Step 3 is intended to identify TECHNOLOGY SOLUTIONS in each sector and these will be target technologies for further analysis*
- ✓ *Step 4 is intended to analyze technology solutions in terms of their characteristics, importance, time of technology realization, and required government policies*



2. Environment Analysis

- Think of what issues/trends might be important in the future of Vietnam for the selected sector
- Collects and analyzes the trends/issues
 - Global trends/issues are gathered from literatures, web sites and by consultation
 - Identify the issues that may have an impact on the future using SWOT analysis for each activity of value chain in the workshop
- Identify key issues using cross impact analysis

(Ex)

❑ *SWOT analysis for each step of value chain*

			Value chain				
			Inputs	Production	Processing	Storage/ Distribution	Markets
SWOT	Internal	Strengths					
		Weaknesses					
	External	Opportunities					
		Threats					

- ❑ *Cross impact analysis to identify key issues*

[illegible]

3. Strategic Products and Services

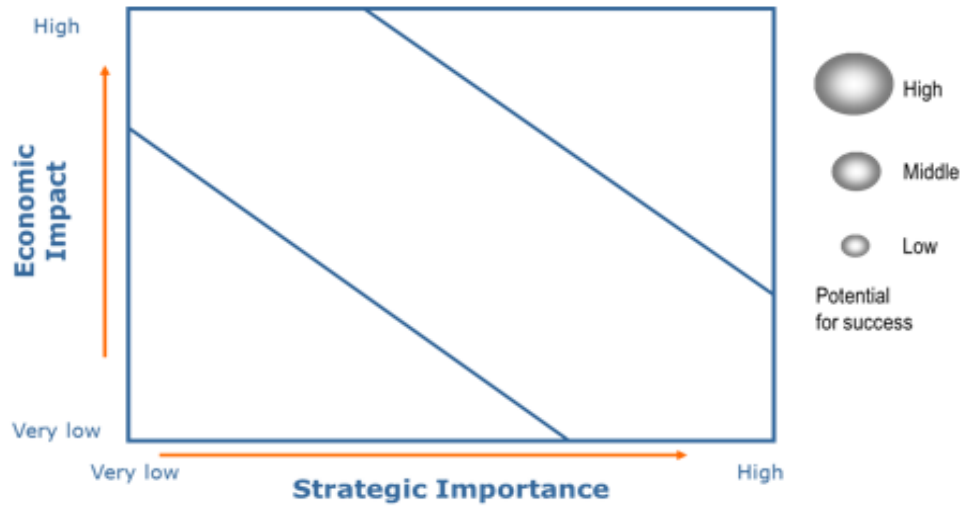
- Identify candidates of strategic products/services to respond to the key issues
- Identify strategic products/services with priority
- (Example) Participants are asked to draw criteria that are important to assess the candidates
- (Example) Participants are asked to assess the candidates in 5-point scale for parameters "economic impact", "strategic importance", and "potential for success"

(Ex)

☐ *Identification of Products/Services*

Value chain	Key issues	Related Actors	Products/Services
A.			
B.			
C.			
D.			
E.			

□ *Prioritization to identify strategic products and services*

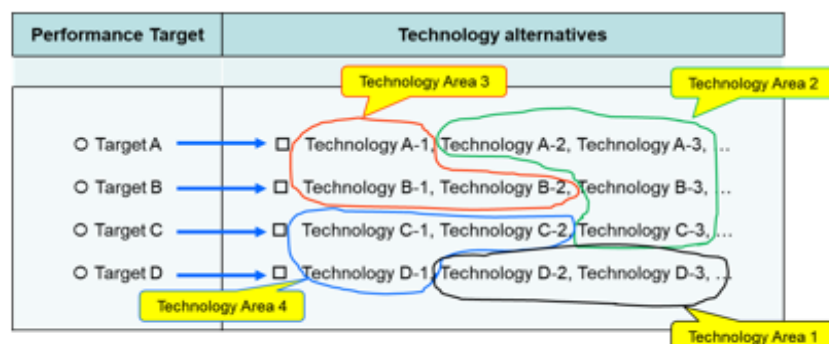


4. Technology Solutions

- Identify performance dimensions that are important for product performance, and that technology can deliver
- The 'Performance' provides a means of linking market, product and technology
- Explore the related technologies to meet the performance dimensions
- Arrange the technologies and develop the technology tree
- Similar technologies will be merged and technologies will be sorted into the appropriate groups

(Ex)

❑ Identification of Technology Alternatives



❑ Technology Tree

Tech Area	Tech Level 1	Tech Level 2

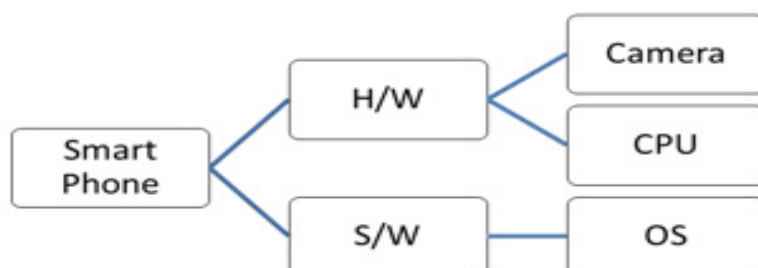


Figure 1. Example of Technology Tree by Component

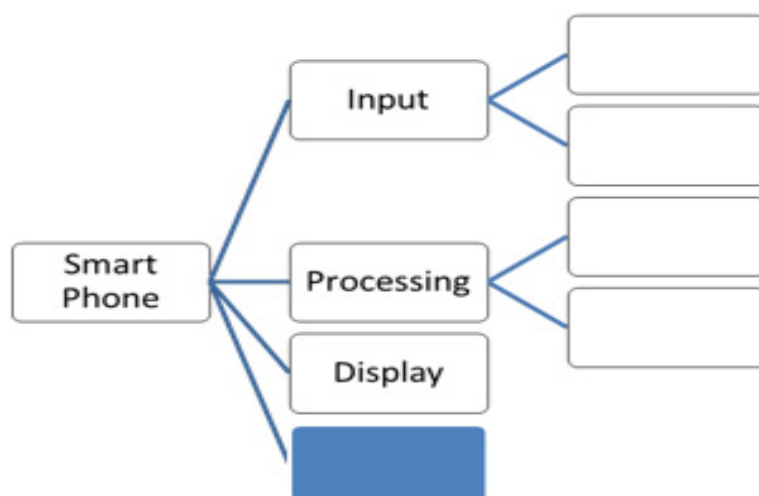


Figure 2. Example of Technology Tree by Functions

5. Technology Analysis

- Analyze technology solutions in terms of their characteristics, importance, time of technology realization, and required government policies



Figure 3. Example of technology analysis

- Each technology is analyzed by 2 round Delphi survey

What is Delphi?

- Delphi is a well-established expert-base technique which recognizes human judgement as legitimate and useful input for generating forecasts and exploring the future impacts of science and technology
- Single experts sometimes suffer biases while panels or group meetings suffer from 'follow the leader' tendencies and reluctance to discard previously stated opinions. These and other weaknesses in the use of expertise have led to the development of Delphi – a sophisticated survey aimed to collect and extract(tacit) knowledge from a group of experts by means of a series of questionnaires
- Questionnaires are normally carried out in rounds or waves (normally two, rarely three)

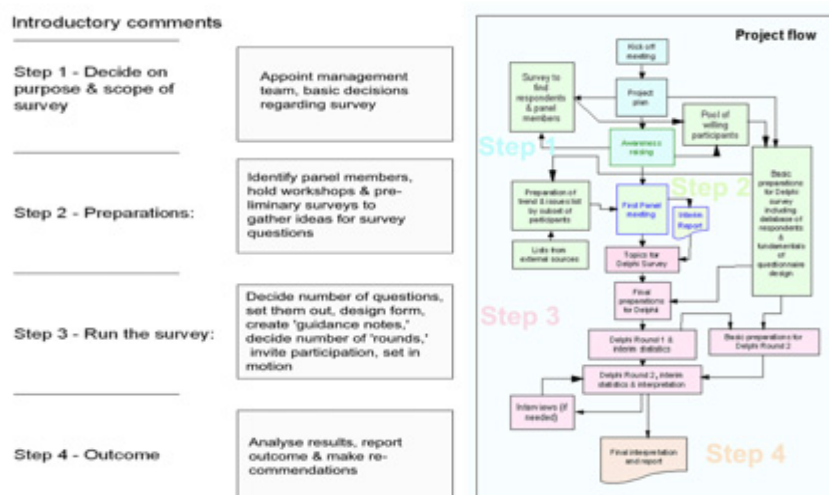


Figure 4. Delphi Process

- Questionnaires in the survey should be clearly and precisely phrased and parameters used in the survey should be comprehensively explained
- Identify key technologies

(Ex)

□ *Delphi survey questionnaire*

Technologies	Realization time				Significance in future society	Technical competitiveness
	'19-'24	'25-'30	'31-'36	> '36		

□ Identification of key technologies

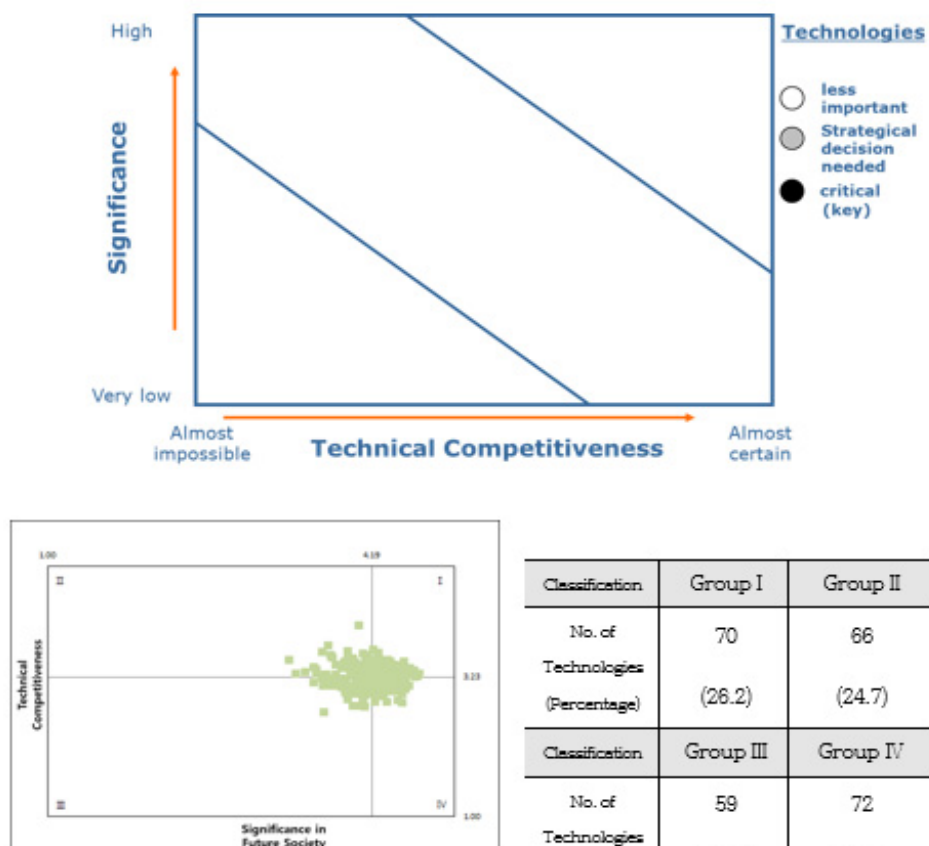


Figure 5. Dispersion of Significance and Technical Competitiveness of Future Technologies in the 5th Science and Technology Foresight

3. Collaboration Process for Capacity Development

3.1 Pilot Project for Vietnam's Technology Foresight

■ Summary

STEPI has prepared to collaborate with VISTI for the pilot project for Technology Foresight in Vietnam. We conducted a pilot project for technology foresight with experts staying in Korea and Vietnam. The results of the pilot project for Vietnam's Technology Foresight have been reviewed in Chapter 3 (see the details in part of "Chapter 3. Sharing Korea's Experience").

- Period: July ~ November 2020
- Purpose: To deliver technology foresight methodologies by promoting Vietnam's technology foresight pilot project

The main items are as follows:

- To set areas of the technology foresight pilot project

Configure survey questions related to traffic in large cities in Vietnam based on 13 promising technologies related to transportation in Vietnam in the future.

- Choose the final 13 promising technologies among 18 related to traffic in Vietnam in the future based on a review by experts.
- Prepare questions including the severity of each issue of traffic, extent of effect of each policy, importance of each promising technology, and period of realization.

[Table 4-6] Survey Contents

Classification	Contents
Basic information	Basic information about the respondents (email, gender, age, occupation, highest educational attainment, etc.)
Severity of traffic issues and policy effect	- Evaluate the severity of each traffic issue in large cities in Vietnam - Evaluate the effect of each policy for solving traffic issues
Promising technology in the future in each area	- Importance of promising technology in 3 areas and common areas - Technical competitiveness in Vietnam - Period of realizing the technology in Vietnam and the world - Core research topics - Common policies needed to realize technology

■ Survey results

1. Recognition of Traffic in Large Cities in Vietnam and Evaluation of Policy Effect

■ Severity of each traffic issue

According to the results of investigation on the severity of each traffic issue in large cities in Vietnam, “parking space issue” got the highest score with 76.7, followed by “road network issue” with 70.6 and “lack of public transportation” with 69.8.

[Table 4-7] Severity Level of Traffic Issue in Large Cities in Vietnam

Classification	Road network issue		Railroad issue		Issue of following traffic laws		Parking space issue		Traffic sign issue		Lack of public transportation	
	Case	Proportion	Case	Proportion	Case	Proportion	Case	Proportion	Case	Proportion	Case	Proportion
Very low	3	5%	5	9%	2	4%	5	9%	2	4%	1	2%
Low	8	15%	9	16%	10	18%	6	11%	12	22%	10	18%
Neutral	16	29%	13	24%	16	29%	4	7%	21	38%	16	29%
High	13	24%	13	24%	18	33%	18	33%	12	22%	17	31%
Very high	15	27%	15	27%	9	16%	22	40%	8	15%	11	20%
Total	55	100%	55	100%	55	100%	55	100%	55	100%	55	100%
Score	70.5		68.7		68.0		76.7		64.4		69.8	

* Score was calculated as very high =100, high=80, neutral=60, low=40, and very low=20.

3.2 Guidelines for National Technology Foresight Exercise

Please check the details of the “Guidelines for National Technology Foresight Exercise” in “Chapter 3. Sharing Korea's Experience” in this report. Below is a summary of the guidelines.

Step 1 is intended to identify key stakeholders and build engagement. They should find and select the experts. They should also organize the committees and assign their roles and responsibilities.

Step 2 is intended to provide new perspectives about the future and identify key issues.

- Identify key issues that affect the future development of Vietnam by scanning the horizon of its current and planned operating and peripheral environments.
- Study the background, future, and potential impacts of these issues.

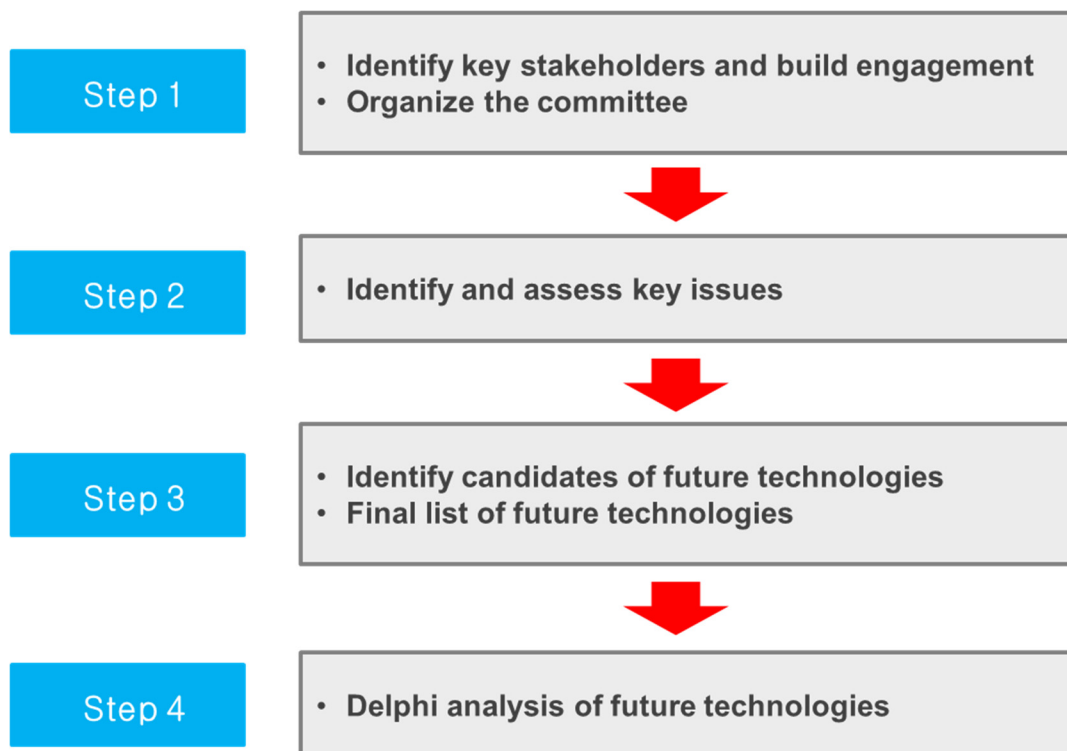
Step 3 is intended to identify future technologies considered to be important in Vietnam to respond to key issues. A brief explanation of each future technology will be given by Vietnamese experts.

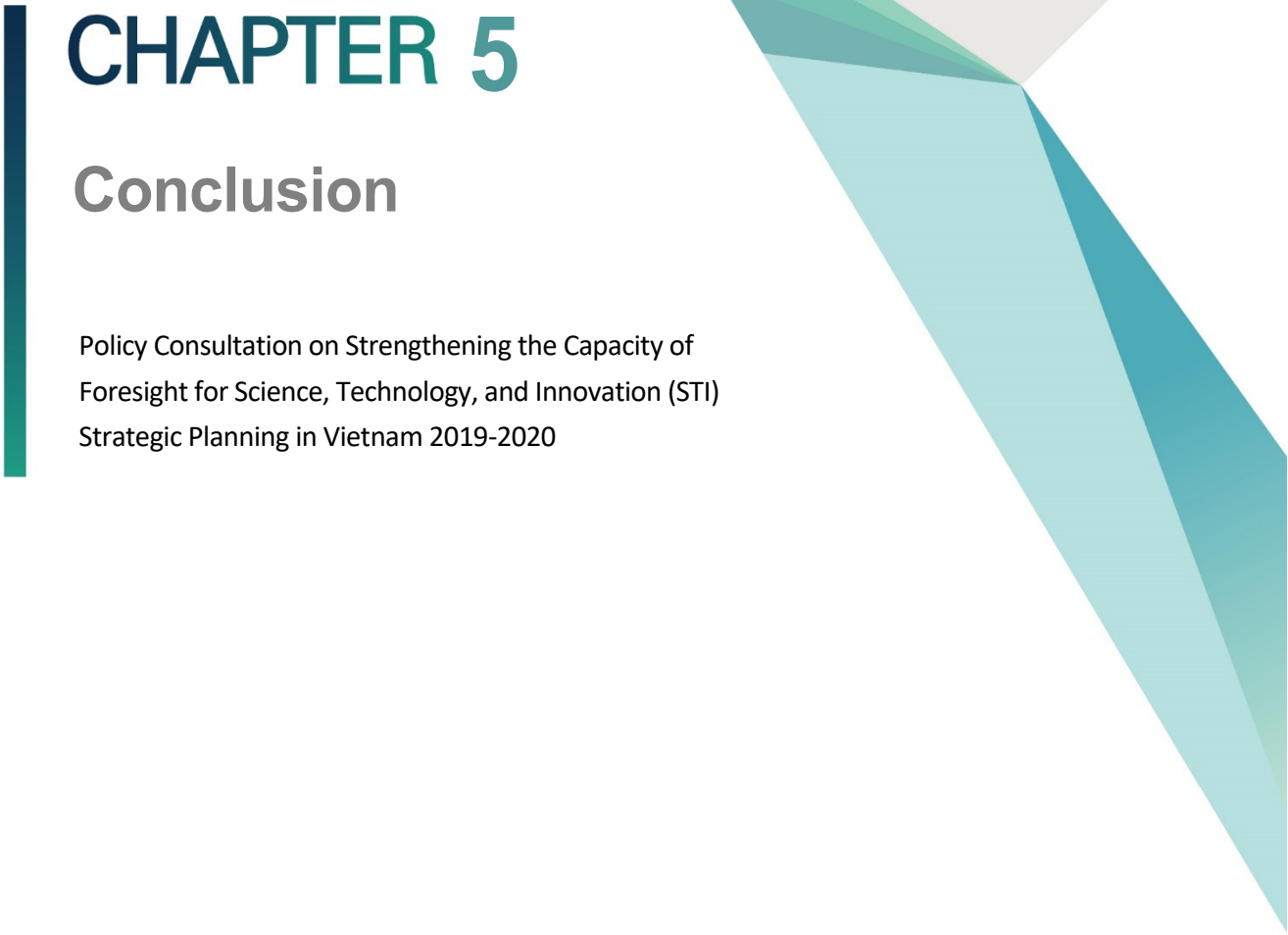
- Prepare an initial list of future technologies to fulfill key issues and needs.
- Search for previously identified future technologies as candidates from other foresights and S&T policies.
- Supplement the future technology list through experts’ consultation.
- Adjust the depth and width of the selected future technologies.
- Draw up the final list of future technologies.
- Provide a brief description of each future technology.

Step 4 is intended to analyze future technologies. A Delphi survey questionnaire will be prepared. Future technologies are analyzed in terms of their characteristics, importance, time of technology realization, and required government policies. It covers the following:

- Design the survey questionnaire.
- Run the survey.
- Analyze the results, report the outcome, and make recommendations.

[Figure 4-4] Process of Technology Foresight Exercise



A decorative vertical bar in dark teal is positioned to the left of the chapter title. To the right, there are abstract geometric shapes: a large light grey triangle pointing left, and several overlapping triangles in various shades of teal and light blue pointing right.

CHAPTER 5

Conclusion

Policy Consultation on Strengthening the Capacity of
Foresight for Science, Technology, and Innovation (STI)
Strategic Planning in Vietnam 2019-2020

Chapter 5. Conclusion

1.

Discussion

■ Serious issues in infrastructure directly related to road traffic

Serious issues in infrastructure directly related to road traffic include “parking space issue,” “road network issue,” and “issue of lack of public transportation” as traffic problems in large cities in Vietnam.

As to the question about the severity of traffic issues, the “parking space issue” (sidewalk or road for most parking space) got the highest score with 76.7, followed by “road network issue” (insufficient connection, concentration, and technical standards of road network) with 70.5 and “issue of lack of public transportation” (qualitative and quantitative lack of public transportation) with 69.8.

On the other hand, the severity related to traffic policies, recognition, and culture such as “issue of insufficient traffic signal” (inefficient traffic control signal system) and “issue of following traffic laws” (road users not properly following traffic laws and insufficient control) was 64.4 and 68.0, respectively. Therefore, these issues were perceived to be less severe than issues related to infrastructure.

■ Need for policies related to road traffic infrastructure

The effect of policies related to infrastructure directly connected with road traffic field such as “improvement of traffic transportation” (score: 74.5) or “road expansion” (score: 73.1) on solving traffic problems turned out to be high.

In contrast, the policy effect on “limit on self-owned vehicles” (score: 60.0), “adjustment of bus line and adoption of exclusive bus line” (score: 61.5), or “establishment of railway network” (score: 63.3) was evaluated as low. This seems to be attributed to the characteristics of Hanoi’s traffic system, which is dominated by private transportation means.

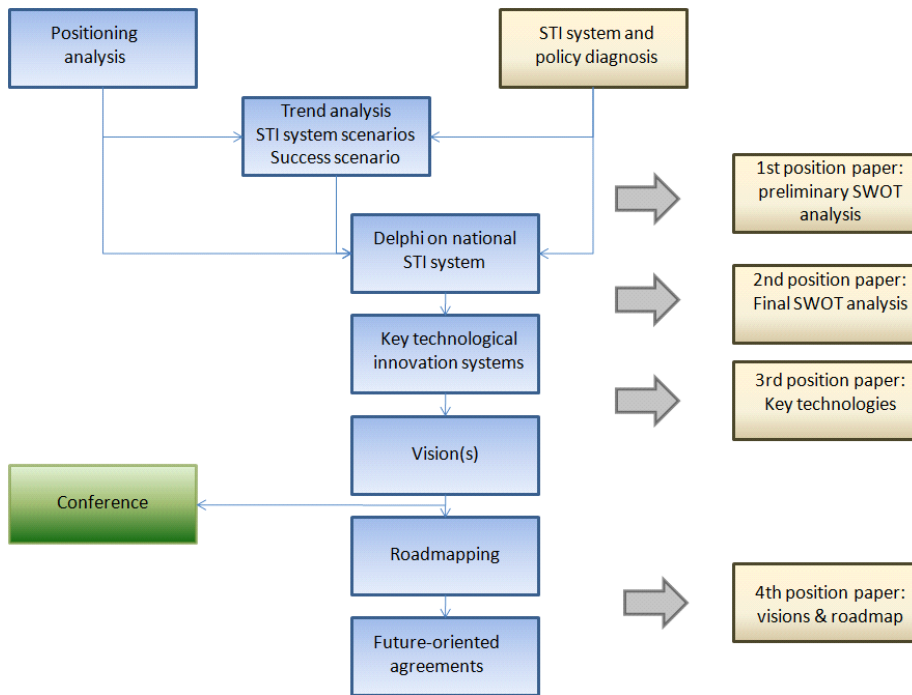
According to statistical data from the Office of Statistics in Vietnam, there were 485,955 vehicles and 5,255,245 motorcycles in 2017. Therefore, the number of vehicles used per 1km of road area is 160, and that of motorcycles is 2,561, suggesting that motorcycles are highly utilized.

2. Implications and Suggestions

This project deduced the guidelines for technology foresight methodology and carried out a pilot project for technical forecasting. Considering the difficult COVID-19 situation, STEPI prepared the “Guidelines for National Technology Foresight Exercise” in lieu of the on-site capacity-building workshop. Such guidelines shed light on the scope of Vietnam’s pilot project of Technology Foresight, methods, and procedures of technology foresight, etc.

Vietnam is a fast-growing developing country, but most found that policy considerations are needed to boost productivity and increase competitiveness for sustainable growth. There are also several crucial challenges that should be tackled such as the polarization, transportation problems, and environmental problems that occur during the rapid growth. Among many policy priorities, continuous investment in and political support for STI are important, since it is a major driving force for economic growth. Our findings on Vietnam’s policy characteristics indicated that a considerable part of Vietnam’s governance adopted the top-down policy design and relatively weak monitoring and evaluation system. Reflecting these characteristics of Vietnam and realizing its future development require establishing national science and technology innovation strategies. As such, technology foresight is an important methodology available in this process.

[Figure 5-1] Procedure for National Innovation Strategy Establishment Based on Technology Forecast



Source: Bastos, C. A. & Weber, M. (2020: 2)

This study also found that there are a few key limitations in Vietnam's science and technology innovation policies. First, during the policy implementation process, monitoring procedures to ensure that participants in the innovation system understand and share strategies are insufficient. Second, there is considerable discord between policy design and policy implementation. This implies the need for a proper monitoring process to ensure strategic policy implementation. Lastly, there is no appropriate evaluation tool for the performance brought about by innovation. Not only financial performance but also innovation capabilities, social capital, and network should be considered.

Technology forecast can be the backbone of strategic tool in addressing these issues and establishing future-oriented innovation policies. Policymakers should consider rapid technological changes and social developments that are difficult to predict. Special attention to and understanding of the impact of these changes and developments and their measurements are required. These results confirm that technology foresight and

technology evaluation are key tools in establishing and implementing STI strategies as the main methodologies for establishing strategies. There appears to be a need to organize and define technology foresight well in government policies.

Lastly, recent emerging patterns in technology foresight include facilitating a dialogue between suppliers and potential users of technology in the technology transfer process. Notably, it may be useful to introduce this facilitation tool in Vietnam, which has introduced and internalized foreign technologies.

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